

LEGO: Synergizing Expert GraphRAG and Expert Chain-of-Thought for Legal Reasoning

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Abstract

Large language models are increasingly applied to high-risk domains such as law, yet complex legal reasoning remains limited by two structural challenges. First, existing RAG and GraphRAG methods emphasize lexical or semantic similarity while overlooking normative relations among legal provisions. Second, vanilla Chain-of-Thought prompting may generate plausible rationales without enforcing the normative structure of legal reasoning. To deal with the bottleneck of pipelines in the legal reasoning domain, we propose **LEGO**, a dual-module framework that synergizes Legal Expert GraphRAG and expert Chain-of-thought for complex legal reasoning. ExpertGraphRAG uses an expert-annotated Civil Code graph encoding these normative relations with a greedy normative-coverage retrieval algorithm to dynamically extract instance-specific provision subgraphs, while ExpertCoT organizes the retrieved provisions and case facts into structured Provision-Fact-Conclusion reasoning. With a Qwen3-8B backbone, LEGO achieves 40.53% exact-match accuracy on LawExamQA_Civil, outperforming the evaluated RAG and CoT baselines and performing comparably to the evaluated larger models, while remaining robust on multi-hop questions. It also achieves the best results among the evaluated baselines on the open-ended benchmarks. Ablation studies confirm the individual and complementary contributions of both modules, demonstrating LEGO’s effectiveness in improving LLMs’ complex legal reasoning ability. Code and dataset can be found in the link: <https://github.com/BLK-WHT/LEGO>

1 Introduction

The integration of LLMs into high-risk professional domains has accelerated rapidly, including law (Chalkidis et al., 2022; Guha et al., 2023; Blair-Stanek et al., 2023). Yet these domains expose a structural gap in two widely used LLM augmentations. Retrieval-Augmented Generation (RAG) grounds generation in external evidence (Lewis et al., 2020; Yu et al., 2022), while Chain-of-Thought (CoT) prompting elicits intermediate rationales (Wei et al., 2022); However, current legal-reasoning applications tend to treat RAG and CoT as isolated components rather than integrating them into a unified reasoning process (Gao et al., 2025).

The first bottleneck is retrieval. Standard RAG and many GraphRAG systems (Edge et al., 2024; Zhang et al., 2025; Zhuang et al., 2026; Chen et al., 2026a; Gutiérrez et al., 2025; Guo et al., 2024) still rely heavily on lexical or embedding similarity, even with induced graph neighborhoods. Yet legal knowledge is semantically sparse: textually distant provisions may be tightly linked through prerequisites, exceptions, general-special relations, or legal effects, while textually similar ones may distract because they belong to different constituent elements. (Yang et al., 2018; Trivedi et al., 2022, 2023). The complex case of Figure 1 shows this issue: repeated transport-related terms may lead similarity-based models to select the transport company (Option D). Yet the Fact Graph shows that B has no contract with the transport company. Correspondingly, the Provision Graph must capture the general-specific relation between contractual privity under Art. 465 and carrier liability under Art. 832, as well as Art.

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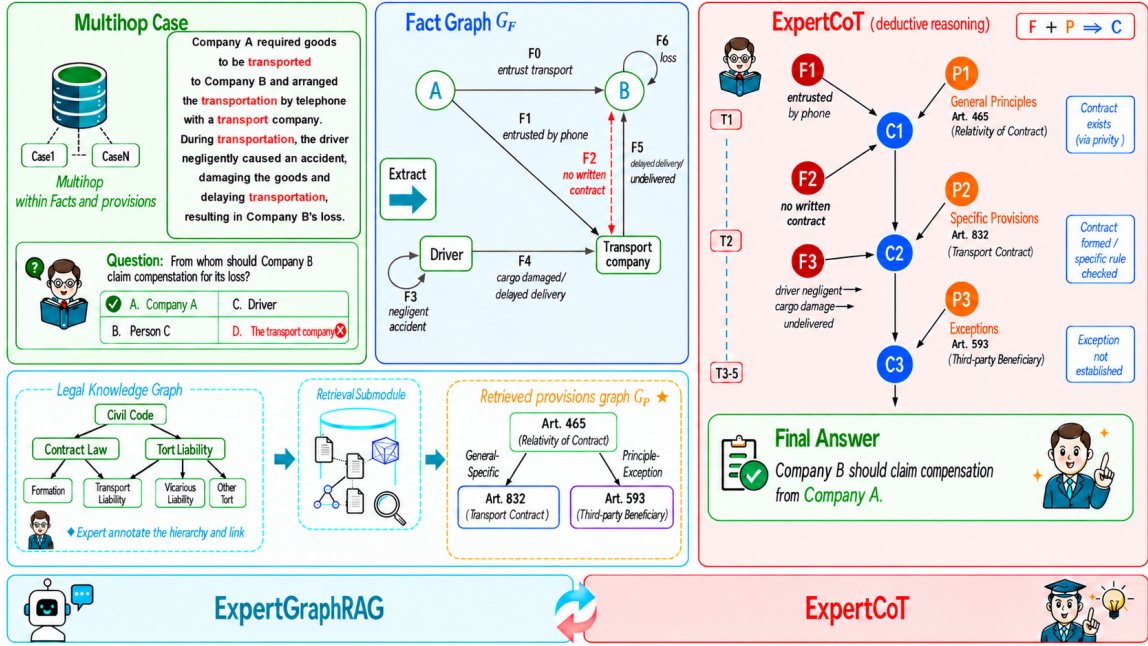


Figure 1: LEGO’s architecture. ExpertGraphRAG uses the fact-side instance representation, implemented as a natural-language query containing the case facts, question, and options to retrieve an instance-specific provision subgraph from the static Expert Provision Graph. ExpertCoT then composes the retrieved provisions and case facts into a structured P–F–C analysis. “Fact Graph” and “Conclusion Graph” in the figure denote conceptual representations rather than separately materialized graph objects.

593’s exclusion of third-party-caused breach as a basis for bypassing contractual privity. Lexical similarity alone captures neither these factual relations nor the normative links among provisions.

The second bottleneck is reasoning. Vanilla CoT and tree-style prompting can produce plausible intermediate text, but they do not guarantee that the model follows the normative order required by a professional domain (Wei et al., 2022; Yao et al., 2023). In law, a reasoning trace can drift by skipping an exception, treating an unfulfilled prerequisite as satisfied, or affirming the consequent from a legal effect back to its condition. Retrieval-augmented reasoning helps only if the retrieved sources are authoritative and structurally relevant; otherwise, each generated step may amplify noise from the previous step, a risk also noted in faithfulness and robustness studies of RAG (Zhou et al., 2026; Gao et al., 2026). Chain-of-Thought (CoT) is extremely fragile and lacks robustness in high-risk domains such as law, where LLM reasoning is easily misled by typos or irrelevant cues (Hu et al., 2025). Research highlights the need to constrain and regulate the models’ CoT using knowledge injection or expert-annotated legal structures.

To resolve the retrieval–reasoning bottlenecks in legal AI and systematically inject expert le-

gal knowledge into the LLM reasoning process, we propose LEGO, a pipeline that incorporates expert legal structure in both retrieval (ExpertGraphRAG) and reasoning (ExpertCoT). First, ExpertGraphRAG retrieves Civil Code articles from an expert-annotated provision graph. Its provision-selection component, *Normative Coverage Greedy* (NCG), selects articles by marginal legal utility rather than isolated semantic similarity, balancing dense relevance, expert-rule alignment, newly covered legal issues, and redundancy control to induce a compact provision subgraph $G_P^{(i)}$ for each input. Second, the expert-guided Syllogistic CoT applies expert constraints to the retrieved provisions and organizes the output as a legal syllogism, producing structured P–F–C reasoning. Experiments and analyses demonstrate LEGO achieves strongest overall empirical performance across legal QA, multi-turn consultation, and practical case analysis on both general and legal-specialized models. It outperforms strong RAG and CoT baselines, remains robust on multi-hop complex cases, and its two expert-guided modules work synergistically to improve legal-reasoning accuracy. The structured syllogistic output also makes the stated legal rationale among provisions, facts, and conclusions easier to inspect and audit.

2 Related Work

2.1 GraphRAG and Legal-Domain Retrieval

Flat-vector RAG can struggle when evidence is distributed across passages because similarity ranking alone does not guarantee retrieval of complete reasoning chains (Han et al., 2026; Trivedi et al., 2023), a challenge isolated by multi-hop QA benchmarks such as HotpotQA, MuSiQue, and 2WikiMultiHopQA (Yang et al., 2018; Trivedi et al., 2022; Ho et al., 2020). GraphRAG instead structures corpus information as graphs and performs graph-aware retrieval or aggregation (Edge et al., 2024; Zhang et al., 2025). Recent work explores efficient or ontology-guided graph construction (Zhuang et al., 2026; Wang et al., 2026), inference-time reasoning structures without pre-built graphs (Chen et al., 2026a), graph-free triplet retrieval (Gong et al., 2026), dependency-aware reranking (Li et al., 2026b), and continual memory (Gutiérrez et al., 2025).

Our RAG baselines include dense-retrieval Naive RAG, graph-memory-based HippoRAG 2 (Gutiérrez et al., 2025), hierarchical-summary-based RAPTOR (Sarathi et al., 2024), subgraph-retrieval-based G-Retriever (He et al., 2024), and entity-relation-based LightRAG (Guo et al., 2024). LegalGraphRAG further organizes legal knowledge into Fact, Ontology, and Rule subgraphs (Chen et al., 2026b). However, neither these general-purpose baselines nor LegalGraphRAG explicitly models typed normative relations among provisions or evaluates retrieval by statute-hop depth.

2.2 Chain-of-Thought and Legal Reasoning

Chain-of-Thought (Wei et al., 2022) and Tree-of-Thought (Yao et al., 2023) prompting externalise intermediate steps but offer no guarantee of normative correctness in vertical domains. Verifier-driven reasoning, popularised by HuatuoGPT-01 (Chen et al., 2024) for medicine, audits each step with a True/False check and triggers backtracking, path-exploration, or self-correction on failure. Legal prompting studies have directly evaluated IRAC-based reasoning on the COLIEE entailment task (Yu et al., 2023) and on expert-annotated legal scenarios (Kang et al., 2023). More recently, MSLR introduced expert-derived IRAC traces for evaluating multi-step legal reasoning (Yu et al., 2025). IRAC-style structured legal reasoning has also been explored on top of ChatLaw (Cui

et al., 2026), Lawformer (Xiao et al., 2021), and LexGLUE (Chalkidis et al., 2022), but these lines of work neither couple reasoning structure with retrieval over an expert-curated graph of normative relations among civil-law provisions nor target civil-law cases that cross multiple statutes.

Recent RAG-CoT systems interleave retrieval with reasoning (Trivedi et al., 2023), revise thought steps with retrieved evidence (Wang et al., 2024), or use knowledge graphs to constrain CoT generation (Li et al., 2025a); expert-guided medical QA further shows that domain attributes can constrain both retrieval and CoT in high-risk settings (Ge et al., 2026). However, Legal reasoning is uniquely challenging because the system must identify and connect the normative relations among multiple provisions and align them with the legally relevant facts to form syllogistic legal reasoning.

3 Methodology

We instantiate LEGO as a pipeline following legal syllogistic reasoning (Patzig, 2013), which decomposes legal inference into the major premise (normative Provision), minor premise (Fact), and conclusion. (Figure 1). For each query q_i , comprising the case facts and question, with answer options O_i , LEGO forms a fact-side query $G_F^{(i)}$. ExpertGraphRAG uses this query to retrieve a set of relevant provisions P_i from the global Provision Graph G_P . ExpertCoT then combines the original query, answer options, and retrieved provision texts to generate a structured P-F-C analysis $G_C^{(i)}$ and the final answer $\hat{A}_i$:

$$\begin{aligned} G_F^{(i)} &:= \text{Serialize}(q_i, O_i), \\ P_i &:= f_{\text{RAG}}\left(G_F^{(i)}, G_P\right), \\ \left(\hat{A}_i, G_C^{(i)}\right) &= f_{\text{CoT}}(q_i, O_i, P_i). \end{aligned} \quad (1)$$

Here, $G_F^{(i)}$ is implemented as the natural-language retrieval query, P_i denotes the retrieved provision texts, and $G_C^{(i)}$ denotes the generated P-F-C response rather than a separately materialized graph. LEGO therefore combines graph-structured provision retrieval with structured syllogistic generation. Unlike prior expert-conditioned QA that uses flat attributes such as subject-area tags (Ge et al., 2026), LEGO represents expert knowledge as a relational legal subgraph and uses CoT as a constrained syllogistic walk over that graph.

3.1 ExpertGraphRAG

LEGO retrieves Civil-Code articles from an offline *Provision Graph*, rather than from free-form chunks. Let $\mathcal{R}$ be expert-defined legal rule units and $\mathcal{A}$ be Civil-Code articles:

$$\begin{aligned} G_P &= (V_P, E_P; \mathbf{B}), \\ V_P &= \mathcal{R} \cup \mathcal{A}, \\ E_P &= E_{\mathcal{R}\mathcal{A}} \cup E_{\text{norm}}. \end{aligned} \quad (2)$$

Here $E_{\mathcal{R}\mathcal{A}}$ is the bipartite rule–article layer, E_{norm} stores expert provision–provision relations, and $B_{r,a}$ is the support weight from article a to rule unit r . For input x_i , let $d_i(a) = \text{sim}(x_i, a)$, $g_i(a) = \max_{r \in \mathcal{R}} \text{sim}(x_i, r) B_{r,a}$, and $\Gamma(a) = \{r : B_{r,a} > 0\}$.

Normative Coverage Greedy. We use *Normative Coverage Greedy* (NCG) to select articles. Given a partial set S , NCG scores a candidate article a by:

$$\begin{aligned} \Delta_i(a \mid S) &= d_i(a) + \lambda_{\text{align}} g_i(a) \\ &\quad + \lambda_{\text{cov}} n_i(a, S) - \lambda_{\text{red}} o_i(a, S), \end{aligned} \quad (3)$$

where $\Gamma(S) = \bigcup_{a \in S} \Gamma(a)$, $n_i(a, S) = |\Gamma(a) \setminus \Gamma(S)|$ is newly covered legal-rule evidence, and $o_i(a, S) = |\Gamma(a) \cap \Gamma(S)|$ is already-covered evidence. The greedy update is:

$$\begin{aligned} a_t &= \arg \max_{a \in \mathcal{A} \setminus S_{t-1}} \Delta_i(a \mid S_{t-1}), \\ S_t &= S_{t-1} \cup \{a_t\}, \quad S_i^* = S_K. \end{aligned} \quad (4)$$

The first three terms are the marginal gain of a relevance-weighted coverage objective over expert rule units. This is the same monotone submodular family underlying maximum coverage and facility-location retrieval, for which greedy selection is the standard approximation strategy (Nemhauser et al., 1978; Lin and Bilmes, 2011). The final term is a lightweight redundancy regularizer: it discourages repeatedly selecting provisions that explain the same legal issue, but we do not use it to claim a new approximation bound. In effect, NCG optimizes the *marginal legal utility* of each provision: it keeps provisions on topic, expands normative coverage, and yields a compact provision subgraph that serves as the legal major-premise context for downstream Syllogistic CoT. The weights λ_{align} , λ_{cov} , λ_{red} and the article budget K used in our experiments are listed in Appendix B.3.

The selected set induces the instance-level provision subgraph:

$$\begin{aligned} G_P^{(i)} &= (V_P^{(i)}, E_P^{(i)}), \\ V_P^{(i)} &= S_i^* \cup \mathcal{R}_i^*, \\ E_P^{(i)} &= E_P \cap (V_P^{(i)} \times V_P^{(i)}). \end{aligned} \quad (5)$$

where $\mathcal{R}_i^* = \{r \in \mathcal{R} : \max_{a \in S_i^*} B_{r,a} > 0\}$.

3.2 ExpertCoT

Let x_i denote the input instance, comprising the case description, question, and answer options. We conceptually represent its fact-side structure as

$$G_F^{(i)} = (V_F^{(i)}, E_F^{(i)}), \quad (6)$$

where $V_F^{(i)}$ denotes the legally relevant entities mentioned in the case and $E_F^{(i)}$ denotes the factual relations expressed among them. This representation is conceptual rather than separately materialized in the implementation. ExpertGraphRAG uses the serialized case, question, and options as its retrieval query, while ExpertCoT jointly generates the P–F–C analysis and final answer from the original instance and the retrieved provisions.

Given the case-level Fact Graph $G_F^{(i)}$ and the retrieved Provision Graph $G_P^{(i)}$, LEGO converts graph-conditioned retrieval into a single-pass structured legal syllogism. Unlike generic legal-analysis prompting, LEGO explicitly simulates deductive legal reasoning by generating the major premise P from the controlling Civil-Code provisions, the minor premise F from the case facts, and the conclusion C from their legally valid composition. Each option-level judgment is therefore represented as a major–minor premise hop: the major premise is one P_i with its typology tag, the minor premise is one or more F_j , and the verdict is in $\{\text{Supported}, \text{Not supported}, \text{Not applicable}\}$. Collecting these hops with the traversed provision subgraph $E_{\downarrow P}^{(i)}$ and the case events from $G_F^{(i)}$ yields the cumulative Conclusion Graph:

$$G_C^{(i)} = (V_F \cup V_P^{(i)} \cup C, E_F \cup E_{\downarrow P}^{(i)} \cup E_C). \quad (7)$$

Here, $V_P^{(i)}$ and $E_{\downarrow P}^{(i)}$ denote the provision nodes retrieved for instance i and their induced relations, rather than the complete global provision graph. where $C = \{\hat{c}_1, \dots, \hat{c}_T\}$ are per-option sub-judgments and E_C records, for each $\hat{c}_t$, the cited P_i , supporting F_j , and any prior $\hat{c}_{t' < t}$ used

as a downstream premise. At fixed top- N retrieval, multi-call P - F - C variants reach similar accuracy on LawExamQA_Civil but are more expensive and harder to audit, so we default to the single-pass variant.

To reduce potential reasoning fallacies and model hallucinations, LEGO adopts not only a syllogistic CoT structure grounded in legal theory, but also a set of explicit behavioral constraints for each component of the syllogism. For the major premise G_P , we impose micro-constraints derived from the structure of the civil-law knowledge graph, requiring the model to distinguish key legal relations, such as general vs. special provisions, general-part vs. specific-part hierarchy, and condition-consequence dependencies. These constraints enable the model to interpret and reason over the latent semantic relations among retrieved provisions rather than relying solely on surface similarity. For the minor premise G_F , the Fact Graph must state only factual predicates without prematurely embedding legal conclusions. For the conclusion G_C , the model may judge each option only after verifying the legal coherence of the provision-fact mapping. Across multiple rounds of expert-in-the-loop error analysis, we iteratively refined these micro-rules to improve the consistency of the generated rationale and make their stated provision-fact-conclusion structure easier to inspect and audit across cases. The full prompt and an end-to-end example are reproduced in Appendix A.

4 Experiments

4.1 Setup

Datasets. We evaluate LEGO on three open datasets in the civil-law domain, ensuring consistency with the expert-constructed civil-law knowledge graph used in our pipeline. (1) **LawExamQA_Civil**, a subset of the Chinese questions from the National Judicial Examination of China (NJE), where we categorize questions by the number of provisions referenced in the official rationale. This hop count serves as a proxy for the complexity of multi-hop legal reasoning. (2) **LexRAG_Civil** (Li et al., 2025b), a civil-law subset of *LexRAG* used to assess multi-turn legal consultation performance. **LexRAG_Civil** evaluates open-ended, multi-turn legal consultation. Models generate free-form responses to successive user queries rather than selecting from predefined answer options. (3) **PLawBench_Civil** (Shi et al.,

2026), a rubric-based benchmark for evaluating LLMs in real-world legal practice. We use the *Practical Case Analysis* section, which consists of complex civil-law cases annotated with reasoning rubrics, enabling assessment of legal reasoning ability in realistic scenarios. Our evaluation is not limited to multiple-choice question answering. Only LawExamQA_Civil uses a multiple-choice format; LexRAG_Civil evaluates free-form responses in multi-turn legal consultations, and PLawBench_Civil evaluates generated practical legal analyses using task-specific rubrics. All of them are evaluated on complex legal-reasoning tasks involving multiple provisions and long case texts. The data details can be found in Appendix D.

Evaluation metrics. For LawExamQA_Civil, we treat the predicted and gold answers as sets of option labels. Accuracy denotes exact set match: a prediction is correct only when the two sets are identical. For each item, we compute set F1 as $F1_i = 2|P_i \cap G_i|/(|P_i| + |G_i|)$, where P_i and G_i are the predicted and gold option sets and report its mean over all items. For example, if $G_i = \{A, B, D\}$ and $P_i = \{A, B\}$, exact match is 0 and set F1 is 0.8. Empty or invalid predictions receive zero for both metrics.

Civil law ExpertGraph. Our expert-annotated Civil Law graph encodes the PRC Civil Code as 318 concepts over 409 rule cards, linked to 1,211 of its 1,260 articles by 4,152 support edges. Each card states one doctrinal rule—its conditions, effects and exceptions—embedded whole, so queries route on rules rather than article text. Built over 7 months by trained legal experts, it serves solely to enhance RAG signal over the civil-code corpus (Appendix F).

LLM Baseline (1) Closed-source and large general LMs. We choose closed-source models GPT-5 (OpenAI, 2025) and DeepSeek-V3 671B (DeepSeek-AI, 2024) as large general-purpose LMs, which have strong zero-shot and long-context abilities, to test the performance ceiling achievable through purely parametric reasoning without external retrieval. (2) Open Larger LMs. We choose Qwen3-30B-A3B (Qwen Team, 2025) and GLM-4.7-Flash (zai-org, 2026) as Open Larger LMs, which offer a strong balance between computational cost and performance at the 30B scale. (3) Small Open-source LMs. We use Qwen3-8B (Qwen Team, 2025) and GLM-4-9B-Chat (zai-org, 2024) as base models

whose 8B–9B parameter scale matches our RAG generators for direct comparison.(4) We select DISC-LawLLM (7B) (Yue et al., 2024) and LegalOne (8B) (Li et al., 2026a), two legal domain finetuned open-source models tailored for legal reasoning.

RAG baseline We compare LEGO with several representative RAG architectures: flat top- k dense retrieval with Qwen3-Embedding-8B; *auto-induced semantic graph* (LightRAG (Guo et al., 2024)); *hierarchical summary tree* (RAPTOR (Sarathi et al., 2024)); *Steiner-tree graph retrieval* (G-Retriever (He et al., 2024)); *long-range memory / associative graph* (HippoRAG2 (Gutiérrez et al., 2025)). All RAG baselines use the same Qwen3-8B backbone and the same Civil-Code retrieval corpus as LEGO, so accuracy differences are attributable to retrieval / reasoning design rather than to scale or external knowledge.

COT baseline Zero-shot-CoT (Kojima et al., 2022) induces models to generate reasoning steps by adding “Let’s think step by step,” while IRAC-CoT is a typical legal reasoning method with Issue–Rule–Application–Conclusion loop demonstrated by MSLR (Yu et al., 2025)

The experiments use the method-specific configurations documented in Appendix B because heterogeneous retrievers define and expand retrieval units differently. These units include chunks, statute articles, PCST seed nodes, tree nodes, and entity–relation candidates, followed by method-specific expansion and serialization procedures. Therefore, no single measure, such as top- k , article count, node count, or token count can fully equalize their retrieval budgets. We accordingly compare complete pipelines under a shared reader, corpus, evaluation set, and decoding configuration, rather than retrieval algorithms using numerically identical but semantically different retrieval units. This follows prior GraphRAG evaluation practice: LegalGraphRAG reports method-specific configurations, while GraphRAG-Bench standardizes top- k where directly applicable and otherwise retains method-specific settings. Details for each baseline are provided in Appendix B.

Metrics. For each item, we compare the predicted answer set against the gold set and report two scores: exact match (EM), which is 1 only when the two sets are identical, and a set-level F1 computed from their overlap. Both are macro-averaged over items rather than over classes; items

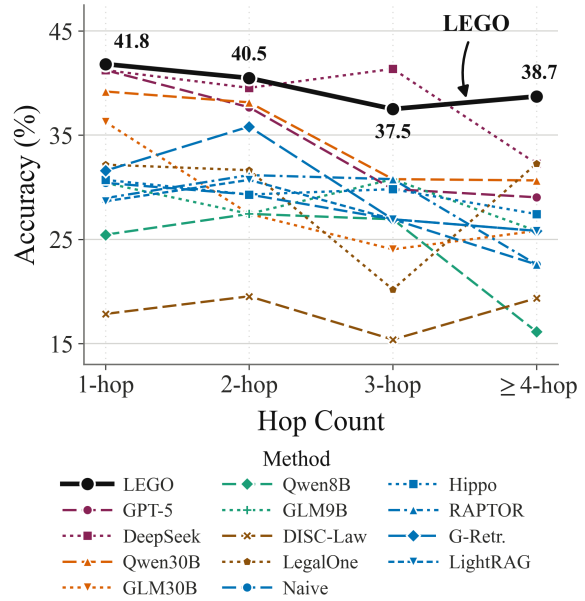


Figure 2: **Multi-hop performance trend.** Accuracy breakdown across hop counts on LawExamQA_Civil.

for which both sets are empty are counted as fully correct (EM = 1, F1 = 1).

We compare LEGO with all 13 systems in Table 1 using two-sided exact McNemar tests on paired per-item predictions, percentile 95% bootstrap confidence intervals from 10,000 item-level resamples, and Holm correction across the complete comparison family.

4.2 Main Results

Table 1 reports the main results on LawExamQA_Civil. LEGO achieves the best overall exact-match accuracy among the evaluated systems, reaching 40.53% with an 8B backbone. This point estimate is numerically higher than those of DeepSeek-V3 (39.97%) and GPT-5 (37.48%), although paired tests do not establish statistically significant superiority over these larger models. By contrast, LEGO significantly outperforms all five evaluated same-backbone RAG systems after Holm correction, including G-Retriever (31.67%; $\Delta = 8.85$ pp, adjusted $p = 4.4 \times 10^{-6}$). The results therefore suggest that external expert structure can compensate for parameter scale, while statistically supported superiority is limited to comparisons under the shared Qwen3-8B backbone and Civil Code corpus. Complete per-comparison statistics are reported in Appendix L.

Hop-robustness and error analysis by case. As shown in Figure 2, LEGO’s advantage is most pro-

Method	Size	LawExamQA_Civil Accuracy (%) by Hop Count				
		Overall	1-hop	2-hop	3-hop	≥4-hop
Closed-source & Large General LMs						
GPT-5	API	37.48	<u>41.23</u>	37.67	29.81	29.03
DeepSeek-V3	671B	<u>39.97</u>	<u>41.23</u>	<u>39.53</u>	41.35	<u>32.26</u>
Open Larger LMs						
Qwen3-30B-A3B	30B	36.93	39.18	38.14	30.77	30.65
GLM-4.7-Flash	30B	30.98	36.26	27.44	24.04	25.81
Small Open-source LMs						
Qwen3-8B	8B	25.45	25.44	27.44	26.92	16.13
GLM-4-9B-chat	9B	29.18	30.41	27.44	30.77	25.81
Legal Domain LMs						
DISC-LawLLM	7B	18.12	17.84	19.53	15.38	19.35
LegalOne	8B	30.29	32.16	31.63	20.19	<u>32.26</u>
RAG Baselines						
Naive RAG	8B	28.91	30.41	29.30	26.92	22.58
HippoRAG 2	8B	29.88	30.70	29.30	29.81	27.42
RAPTOR	8B	29.32	28.95	31.16	30.77	22.58
G-Retriever	8B	31.67	31.58	35.81	26.92	25.81
LightRAG	8B	28.77	28.65	30.70	26.92	25.81
LEGO(ours)	8B	40.53	41.81	40.47	<u>37.50</u>	38.71

Table 1: **Main results stratified by multi-hop reasoning complexity on LawExamQA_Civil.** Performance is measured by exact-match accuracy (%). **Bold** indicates the absolute best performance across all evaluated models, while underline indicates the second-best. By integrating ExpertGraphRAG and ExpertCoT, the 8B-parameter LEGO achieves the highest observed overall accuracy and exhibits remarkable robustness in deep reasoning chains (≥ 4 -hop).

nounced on deeper reasoning cases. On ≥ 4 -hop questions, LEGO reaches 38.71%, outperforming the second-best result of 32.26% by 6.45 points. It also shows stronger hop-resilience: accuracy remains relatively stable across 1/2/3/4+-hop items (41.8 $\rightarrow$ 40.5 $\rightarrow$ 37.5 $\rightarrow$ 38.7), while GPT-5 drops by 12.2 points from 1-hop to 4+-hop cases the five RAG baselines drop by 2.8–7.8 points. To diagnose this robustness, we analyze the items correctly answered by LEGO but missed by all 13 baselines. Four recurring patterns emerge: general-vs-special rule resolution, doctrinal selection between adjacent liability regimes, threshold-gated claims requiring constitutive-element checks, and right-holder identification in multi-party settings. These four patterns are a qualitative taxonomy; they should not be confused with the twenty cases in Appendix H, which are defined by a deliberately stringent fourteen-system intersection (LEGO correct and all thirteen alternatives wrong). These patterns correspond closely to the priority, prerequisite, exception, and entitlement relations encoded in the Civil Law ExpertGraph, suggesting that LEGO’s

gains come from exploiting normative legal structure rather than surface-level retrieval alone. Appendix H expands one representative case per pattern.

LexRAG_Civil and PLawBench_Civil. As shown in Table 2, LEGO achieves the highest among all RAG baselines on both benchmarks. On LexRAG_Civil, LEGO achieves the best score in every evaluated dimension, including factuality (4.486), satisfaction (4.600), clarity (6.879), coherence (5.700), completeness (5.093), and overall quality (5.164). Compared with the strongest baseline, HippoRAG 2, LEGO improves the overall score by 0.335 points and completeness by 0.407 points. On PLawBench_Civil, LEGO obtains the highest reasoning score (47.40) and overall score (59.68), suggesting that its expert-structured retrieval and reasoning framework benefits long-form legal reasoning.

Beyond aggregate scores, our error analysis shows that LEGO mainly reduces three recurring failures of generic RAG systems: retrieving lexically similar but legally irrelevant provisions, pro-

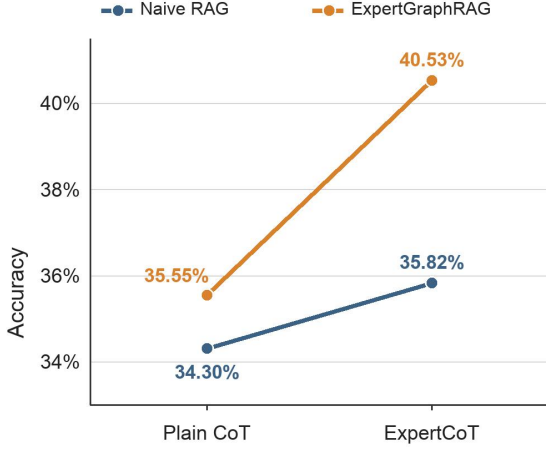


Figure 3: **Interaction between ExpertGraph and Syllogistic CoT.** Accuracy improves most when both components are enabled.

ducing partial answers based on a single provision, and giving hedged conclusions when the legal issue requires a determinate judgment. By combining ExpertGraphRAG with Syllogistic CoT, LEGO retrieves structurally relevant provisions, composes them into more complete reasoning chains, and produces clearer final judgments. These results demonstrate that LEGO improves not only subjective answer quality, but also the robustness and practical usability of legal reasoning in multi-turn and long-form civil-law QA.

4.3 Ablation Study

Table 3 shows that the full LEGO system consistently outperforms all baselines and variants in both Accuracy and F1, validating the contribution of each core component. When the RAG module is removed, or when the model relies only on standard prompting strategies such as Qwen3-8B zero-shot, CoT, and IRAC-CoT, performance drops substantially. Compared with these No-RAG baselines, LEGO improves Accuracy by 9.41–15.08 percentage points (pp) and F1 by 3.05–9.59 pp, highlighting the difficulty of complex legal reasoning without external legal knowledge retrieval. Even with retrieval augmentation, general RAG frameworks paired with standard prompts, such as Naive RAG and G-Retriever with CoT/IRAC-CoT, still lag behind the full system by 6.23–8.30 pp in Accuracy and 3.56–6.13 pp in F1. This suggests that generic retrieval pipelines struggle to capture the rigorous structure required for statutory application. Moreover, although RAG + ExpertCoT variants benefit

from expert-annotated reasoning guidance, they remain 4.71–4.98 pp lower in Accuracy and 2.72–4.07 pp lower in F1 than LEGO, underscoring the importance of our structurally optimized ExpertGraphRAG for reducing semantic noise and supporting deeper reasoning chains. Figure 3 further shows that the benefit of each component is larger when the other is present: ExpertGraphRAG improves Accuracy by 1.24 pp with Plain CoT but by 4.70 pp with ExpertCoT, while ExpertCoT yields gains of 1.52 and 4.98 pp with Naive RAG and ExpertGraphRAG, respectively. The corresponding interaction analysis is reported in Appendix L.

4.4 Recall of Legal Retrieval

Table 4 shows that ExpertGraph substantially improves gold-context approximation over plain retrieval. Compared with Naive RAG, CivilCodeRAG + ExpertGraph raises Recall@8 from 27.99% to 72.53%, Recall@10 from 34.70% to 74.58%, and Recall@20 from 50.52% to 81.07%. This improved provision coverage also translates into stronger downstream QA performance, increasing accuracy from 25.45% in the zero-shot setting to 30.29%. Notably, ExpertGraph achieves substantially higher coverage of gold provisions while relying entirely on automatically retrieved context, indicating that structured legal-graph expansion helps recover relevant provisions missed by plain BM25 or dense retrieval. To diagnose why ExpertGraph helps, Table 4 decomposes retrieval quality into four facets and reports the **Gap Closing Rate**

$$\text{GCR}(M) = \frac{\text{ACC}(M) - \text{ACC}_{\text{zeroshot}}}{\text{ACC}_{\text{gold}} - \text{ACC}_{\text{zeroshot}}} \quad (8)$$

LEGO performs close to the gold-article reference. The ExpertGraph is built once, at corpus level, independently of the benchmark questions, answer labels and official rationales, and is then reused across downstream legal reasoning tasks without any task-specific annotation of gold provisions. Under that setting LEGO reaches 40.53% accuracy, against 41.22% for the gold-article reference in Table 3 (Gold Article + ExpertCoT), which is handed the controlling provisions directly. The 0.69 pp gap corresponds to five items out of 723: automatically retrieved provisions recover essentially all of the benefit of manually supplied ones. Constructing the ExpertGraph requires non-trivial upfront effort, but because that effort is corpus-level rather than item-level, it is amortised across

Benchmark	Metric	Naive RAG	RAPTOR	G-Retriever	HippoRAG 2	LightRAG	LEGO
LexRAG_Civil	Factuality	3.707	3.564	3.829	<u>4.329</u>	4.121	4.486
	Satisfaction	3.829	3.814	3.986	<u>4.300</u>	4.043	4.600
	Clarity	6.029	6.029	6.136	<u>6.586</u>	6.286	6.879
	Coherence	4.814	4.743	4.921	<u>5.329</u>	5.257	5.700
	Completeness	4.193	4.129	4.307	<u>4.686</u>	4.279	5.093
	Overall	4.314	4.293	4.450	<u>4.829</u>	4.543	5.164
PLawBench_Civil	Reasoning	<u>46.50</u>	44.70	43.00	45.30	43.53	47.40
	Overall	<u>59.05</u>	58.02	57.22	57.93	58.43	59.68

Table 2: Cross-benchmark evaluation on LexRAG_Civil and PLawBench_Civil.

Method	Context Coverage (%)			QA		GCR
	Recall@8	Recall@10	Recall@20	Acc (%)	F1	
Naive RAG	27.99	34.70	50.52	28.91	52.16	0.61
BM25 (CivilCode, plain)	45.55	48.21	57.90	29.05	52.96	0.64
Dense (CivilCode, plain)	68.36	71.93	79.19	29.60	55.07	0.73
LEGO (CivilCode,w/ ExpertGraph)	72.53	74.58	81.07	30.29	54.78	0.85
Gold Article	100	100	100	31.12	55.41	1.00

Table 4: **Gold Context Approximation on LawExamQA_Civil.** Coverage is reported as Recall@ k at $k \in \{8, 10, 20\}$. GCR = Gap Closing Rate on QA accuracy.

Model	Description	LEGO		Improvement (Δ)	
		Acc	F1	Δ Acc	Δ F1
No-RAG Baselines					
Qwen3-8B	zero-shot	0.2545	0.5382	$\uparrow 0.1508$	$\uparrow 0.0305$
Qwen3-8B	CoT	0.2918	0.4728	$\uparrow 0.1135$	$\uparrow 0.0959$
Qwen3-8B	IRAC-CoT	0.3112	0.4973	$\uparrow 0.0941$	$\uparrow 0.0714$
RAG + Standard CoT					
Qwen3-8B	Naive RAG + CoT	0.3430	0.5159	$\uparrow 0.0623$	$\uparrow 0.0528$
Qwen3-8B	Naive RAG + IRAC-CoT	0.3223	0.5312	$\uparrow 0.0830$	$\uparrow 0.0375$
Qwen3-8B	G-Retriever + CoT	0.3375	0.5074	$\uparrow 0.0678$	$\uparrow 0.0613$
Qwen3-8B	G-Retriever + IRAC-CoT	0.3347	0.5331	$\uparrow 0.0706$	$\uparrow 0.0356$
RAG + ExpertCoT					
Qwen3-8B	Naive RAG + ExpertCoT	0.3582	0.5280	$\uparrow 0.0471$	$\uparrow 0.0407$
Qwen3-8B	G-Retriever + ExpertCoT	0.3555	0.5415	$\uparrow 0.0498$	$\uparrow 0.0272$
ExpertGraphRAG + Standard CoT					
LEGO	CoT	0.3555	0.5312	$\uparrow 0.0498$	$\uparrow 0.0375$
LEGO	IRAC-CoT	0.3734	0.5456	$\uparrow 0.0319$	$\uparrow 0.0231$
ExpertGraphRAG + ExpertCoT					
LEGO	Full System	0.4053	0.5687	-	-
Gold Article					
Gold Article	+ CoT	0.3472	0.5257	-	-
Gold Article	+ IRAC-CoT	0.3790	0.5468	-	-
Gold Article	+ ExpertCoT	0.4122	0.5894	-	-

Table 3: **End-to-end RAG performance and ablation study.** Each row reports the full LEGO system’s Δ Acc/ Δ F1 over that configuration.

additional datasets and tasks rather than repaid for each new benchmark.

We defer the full error analysis of the 430 mis-answered items to Appendix G, including the error-type distribution (Figure 8) and the full table (Table 18).

5 Conclusion

To address the robustness limitations of large language models in complex legal reasoning, we present **LEGO**, a framework that integrates ExpertGraphRAG and ExpertCoT for the legal domain. ExpertGraphRAG is a retrieval-augmented generation method built on an expert-constructed legal knowledge graph, which improves grounding, preserves the structural relations of legal knowledge, and supports accurate identification and linking of relevant provisions. ExpertCoT further incorporates expert legal reasoning to interpret retrieved provisions and enhance the accuracy and consistency of multi-hop reasoning. Our evaluation demonstrates that combining an expert-knowledge-grounded GraphRAG with expert-guided chain-of-thought reasoning yields synergistic improvements in the robustness of complex legal reasoning for large language models.

Limitations

This paper focuses on a methodological framework rather than proposing a new legal reasoning benchmark. LawExamQA_Civil is used only as an evaluation set to validate expert-structured retrieval and syllogistic reasoning, not as a standalone public multi-hop benchmark. Its hop count is approximated by the number of Civil Code provisions cited in official rationales, which does not yet capture fine-grained inter-provision transitions such as prerequisite, exception, priority, condition–consequence, or general–special relations. Our experiments evaluate answer accuracy, but do not establish either the faithfulness of the generated analyses to the model’s internal reasoning process or their correctness as post-hoc explanations. Moreover, although the Civil Law Expert-Graph is constructed independently of benchmark questions, answer labels, and official rationales, building and maintaining such a normative graph requires substantial expert effort and may limit scalability across jurisdictions and legal domains. The Fact Graph is produced via single-pass LLM extraction with limited expert checks and is therefore not a fully reliable factual benchmark.

Ethics Statement

LEGO is a methodological research framework for evaluating expert-structured retrieval and syllogistic reasoning in legal question answering. It is not intended to provide legal advice, replace qualified legal professionals, or support automated legal decision-making. All evaluation questions are derived from publicly available judicial-examination materials and are used only for controlled experimental evaluation. This work does not propose LawExamQA_Civil as a new public benchmark, nor does the evaluation encode or promote any particular legal interpretation. The Civil Law Expert-Graph is constructed independently of benchmark questions, answer labels, and official rationales. Expert annotators were compensated at fair market rates.

Use of AI Assistants

We acknowledge the use of several large language models, including Claude, Gemini, GPT, and Qwen, as well as GitHub Copilot, to assist with text editing and code development during this research. All outputs were thoroughly verified by

the authors, who maintain full accountability for the paper’s contents.

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A Syllogistic CoT Prompt Card

This appendix reproduces the full prompts used by LEGO’s Syllogistic CoT step (§3.2) on the production run that obtains 0.4053/0.5687 on LawExamQA_Civil. Both the SYSTEM and USER prompts are passed to the same Qwen3-8B backbone in a single call with greedy decoding (temperature = 0.0, max_tokens = 1800).

A.1.1 System prompt

PROMPT: *You are a careful multiple-choice assistant for PRC civil-law questions. You must reason strictly from the case facts, the options, and the provided statutes. Do not invent facts or statutes that are not given.*

A.1.2 User prompt template

The user prompt is a fixed template into which three blocks are filled at run-time: {law_context} (the top-*N* retrieved Civil-Code articles, each rendered as “[k] PRC Civil Code Art. c: ...”); {case} (the case body); and {question_with_options} (the question stem followed by the four options A–D).

PROMPT:

Please solve this PRC civil-law multiple-choice question using the P/F/C (Provision–Fact–Conclusion) method.

Role: write as a neutral adjudicator / exam grader.

Requirements:

1. In one sentence, state the legal object to be decided and what the question asks.
2. P-Provision: list only the controlling rules among the [Provided Statutes]; do not add any new statutes.
3. In P, explicitly state the provision relations (general vs. special rule, proviso/exception, limiting condition, remedy) and which one controls the conclusion.
4. Carefully distinguish: contract validity, real-rights transfer, opposability, liability allocation, damages, priority rights / defenses; do not expand one legal effect into another.
5. When you see limiting clauses (e.g., “shall not be deemed invalid solely because...”, “does not affect the validity of...”, “may claim compensation”), interpret them as limitations; do not reverse-infer invalidity or automatic extinction of rights.
6. F-Fact: list only the legally relevant facts; do not write the final legal conclusion.
7. C-Conclusion: first perform a “provision-relation self-check”, then judge options A/B/C/D. Use at most one sentence per

option.

8. Select options according to the question polarity. If it asks for “incorrect / unlawful / not established”, choose the incorrect option(s).
9. Do not multi-select for coverage. Select an option only when its subject, legal predicate, required elements, and claimed legal effect all match the Provision–Fact mapping.
10. Keep each section to 1–3 sentences; focus on comparing options rather than repeating the case.
11. The last line must be exactly [FINAL_ANSWER]X<eo>, e.g., [FINAL_ANSWER]A<eo> or [FINAL_ANSWER]BD<eo>.

[Provided Statutes]

{law_context}

[Case]

{case}

[Question/Options]

{question_with_options}

Output format:

Object:

P-Provision:

Provision relations:

F-Fact:

C-Conclusion:

Provision-relation self-check:

[FINAL_ANSWER]...<eo>

A.1.3 End-to-end example (qid 0)

The following is the user prompt as actually sent for question qid 0 (2002 Paper III No.1, the transport-delay / cargo-damage case used as the worked example in Appendix C). Only the three slot fillers ({law_context}, {case}, {question_with_options}) differ from the template above.

{law_context}.

{law_context}:

[1] *PRC Civil Code* Art. 832: The carrier shall be liable for compensation for any damage to or loss of the goods during transport. However, where the carrier proves that such damage or loss was caused by force majeure, the natural properties of the goods or reasonable

wear and tear, or the fault of the consignor or consignee, the carrier shall not be liable for compensation.

[2] *PRC Civil Code* Art. 834: Where two or more carriers undertake through-transport by the same mode of transport, the carrier that entered into the contract with the consignor shall be liable for the whole course of transport; where the loss occurs on a particular segment, the carrier that entered into the contract with the consignor and the carrier responsible for that segment shall be jointly and severally liable.

[3] *PRC Civil Code* Art. 825: Where the consignor handles the carriage of goods, it shall accurately state to the carrier the name of the consignee, the name or the consignee as instructed, and the necessary information about the goods for carriage such as the name, nature, weight, quantity, and place of delivery. Where the consignor makes a false declaration or omits important information and thereby causes losses to the carrier, the consignor shall be liable for compensation.

[4–8] *PRC Civil Code* Arts. 841 / 842 / 824 / 607 / 837 (multimodal transport, passenger transport, risk allocation in sales contracts, deposit/consignation, etc.; full text omitted; not on the main reasoning path of this question).

{case}.

{case}:

Company A needs to ship a batch of goods to Company B (the consignee). The parties agreed that Company A would arrange the logistics. Company A's legal representative, C, contacted by phone and commissioned a trucking company holding a road-transport business license to transport the goods. The trucking company assigned its full-time employee driver, Liu, to drive a dedicated vehicle. During transport, a traffic accident occurred due to Liu's negligence; the traffic police accident report found Liu fully at fault, and the goods were damaged. Company B had previously queried the trucking company directly about the location of the goods. Company B suffered losses because it failed to receive the goods in time.

{question_with_options}.

{question_with_options}:

Since the damage was directly caused by the transport side, who should Company B claim compensation from?

Options:

A. Company A

B. C

C. Liu

D. The trucking company, as the actual carrier that directly caused the damage; holding it liable for damages accords with tort-law principles.

Expected output skeleton. The model returns the five labelled sections of §3.2 populated against this input, ending with the token [FINAL_ANSWER]A<eo> (the gold answer for this item). The full P / F / C body is stored in the prediction log and is the source for the worked example in Appendix C.

B Baseline Details and Hyperparameters

This section lists the configurations used for baselines reported in Table 1, Table 2, and Table 3.

B.1 RAG Baselines

Naive RAG. Naive RAG (Lewis et al., 2020) retrieves the top- k Civil-Code chunks by dense similarity and concatenates them as context for the generator.

Naive RAG Configuration

```
{
  embedding_model: Qwen3-Embedding-8B,
  retrieval_topk: 5,
  chunk_token_size: 1000,
  chunk_overlap_token_size: 200
}
```

G-Retriever. G-Retriever (He et al., 2024) solves a Prize-Collecting Steiner Tree over the Civil-Code knowledge graph, returning a connected sub-tree spanning the top- k prize nodes and their relations.

G-Retriever Configuration

```
{
  embedding_model: Qwen3-Embedding-8B,
  retrieval_topk: 10,
  chunk_token_size: 1200,
  chunk_overlap_token_size: 100,
  entities_max_tokens: 3000,
  relationships_max_tokens: 2000
}
```

LightRAG. LightRAG (Guo et al., 2024) performs hybrid retrieval over an auto-induced entity-relation graph, scoring queries against a global-relation channel and a local-entity channel with separate token budgets.

LightRAG Configuration

```
{
  embedding_model: Qwen3-Embedding-8B,
  query_type: hybrid,
  retrieval_topk: 20,
  chunk_token_size: 1200,
  chunk_overlap_token_size: 100,
  max_token_global_context: 2000,
  max_token_local_context: 2000,
  max_token_text_unit: 2000
}
```

HippoRAG 2. HippoRAG 2 (Gutiérrez et al., 2025) links query mentions into a pre-built fact / passage hybrid graph and accumulates evidence via multi-step traversal before returning the final passages.

HippoRAG 2 Configuration

```
{
  embedding_model: Qwen3-Embedding-8B,
  retrieval_top_k: 10,
  linking_top_k: 7,
  qa_top_k: 10,
  max_qa_steps: 3,
  graph_type: facts_and_sim_passage_
    node_unidirectional
}
```

RAPTOR. RAPTOR (Sarathi et al., 2024) constructs a hierarchical summary tree by recursively clustering and summarising chunks, enabling retrieval at multiple levels of abstraction; we use the collapsed-tree variant.

RAPTOR Configuration

```
{
  embedding_model: Qwen3-Embedding-8B,
  retrieval_topk: 10,
  chunk_token_size: 1200,
  chunk_overlap_token_size: 100,
  num_layers: 5,
  max_length_in_cluster: 3500,
  threshold: 0.1,
  cluster_metric: cosine,
  threshold_cluster_num: 5000,
  max_context_tokens: 4500
}
```

B.2 CoT Baselines

CoT baselines pair with any retriever in the ablation: the retrieved articles fill the law_context block while the scaffold controls how the generator consumes them.

Zero-shot CoT. Zero-shot CoT (Kojima et al., 2022) prepends “Let’s think step by step” and lets the model freely generate a reasoning chain before committing an answer.

Zero-shot CoT Configuration

```
{
  backbone: Qwen3-8B,
  sampling: greedy,
  max_tokens: 1800,
  top_n_articles: 8,
  scaffold: "Let's think step by step",
  output_skeleton: [Reasoning] →
    [Option check] →
    [Answer]X<ea>
}
```

IRAC-CoT. IRAC-CoT (Yu et al., 2025) forces a four-section Issue–Rule–Application–Conclusion trace before the answer token.

IRAC-CoT Configuration

```
{
  backbone: Qwen3-8B,
  sampling: greedy,
  max_tokens: 1800,
  top_n_articles: 8,
  scaffold: Issue → Rule →
    Application → Conclusion,
  output_skeleton: [Issue] → [Rule] →
    [Application] →
    [Conclusion] →
    [Answer]X<eo>
}
```

B.3 Experiment Settings

For LEGO and all retrieval-based baselines, we use Qwen3-8B as the generator to ensure a fair backbone-controlled comparison. During generation, all open-source experiments use greedy decoding with temperature=0 and max_tokens=1800 (512 for the closed-book zero-shot setting, whose output is a bare answer tag); the Qwen3-8B reader is run with thinking mode disabled (enable_thinking=False) in all configurations; API-based experiments use the providers’ default decoding configurations. For LEGO’s provision retrieval on LawExamQA_Civil, we instantiate the Normative Coverage Greedy (NCG) objective of Eq. 3–4 with the weights listed below; all d_i / g_i similarities are min–max normalized to $[0, 1]$ per query before the greedy selection. To guarantee reproducibility, we fix the random seed at 42. All open-source model inferences and evaluations are conducted on a single server equipped with $8 \times$ NVIDIA H100 (80GB HBM3) GPUs.

LEGO Configuration

```
{
  backbone: Qwen3-8B,
  embedding_model: Qwen3-Embedding-8B,
  sampling: greedy,
  temperature: 0.0,
  max_tokens: 1800,
  seed: 42,
   $\lambda_{\text{align}}$ : 0.55, // Eq. 3
   $\lambda_{\text{cov}}$ : 0.18, // Eq. 3
   $\lambda_{\text{red}}$ : 0.06, // Eq. 3
  article_budget K: 8, // Eq. 4
  top_n_articles N: 8
}
```

The three weights are lightly tuned on a held-out development split, giving the largest weight to query–article alignment ($\lambda_{\text{align}} = 0.55$), a moderate weight to marginal normative-rule coverage ($\lambda_{\text{cov}} = 0.18$), and a small redundancy penalty ($\lambda_{\text{red}} = 0.06$). The article budget $K = N = 8$ means NCG selects eight articles per query and all

eight are passed into the Syllogistic-CoT prompt as {law_context}.

C P/F/C Case Studies with Pipeline-Level Reasoning

This appendix expands six representative ablation examples into case-level P/F/C (Provision–Fact–Conclusion) comparisons. The original benchmark cases are Chinese civil-law multiple-choice questions; to avoid encoding problems in the appendix, the case facts and options below are English renderings of the original items. For each case, green cells mark legally correct P/F/C steps, red cells mark the step that causes or directly contributes to the wrong answer, and yellow cells mark partially correct reasoning that still misses the decisive legal boundary.

C.1 QID 19: Contract Rights and Obligations Transfer

Case. Company A and Korean Company B formed a Chinese-foreign joint venture, and the joint-venture contract had already been approved by the competent authority. Company C is a large state-owned enterprise with the relevant qualifications. Company A wrote to Company B: “We plan to transfer our equity to Company C. If you object, please raise a written objection within 30 days; otherwise you will be deemed to have consented.” Company A then transferred its contractual rights and obligations to Company C without Company B’s consent. Company B did not reply in writing for three months after receiving the notice. Company C had already participated in the joint venture’s operations for one year, with good business results, and Company B had once signed a board resolution together with Company C.

Question and options. Given that Company C has actually performed contractual obligations and Company B did not object in time, is the transfer of the contract illegal?

- A. Yes.
- B. No. Based on commercial efficiency and actual performance, the transfer should be treated as valid, and Company B impliedly recognized it by joining board actions.
- C. It is illegal only if Company B expressly objects.

Observation	QID	Gold	Prediction contrast	Dominant P/F/C difference
Obs. 1 retrieval helps	19	A	zero-shot B vs. RAG/LEGO A	P: retrieved transfer-consent rules block implied-consent intuition.
Obs. 1 retrieval insufficient	4	D	plain RAG B vs. LEGO Full D	P: apparent agency must override bare no-authority reasoning.
Obs. 2 Expert-CoT at fixed retrieval	1	C	CoT CD vs. ExpertCoT C	C: termination damages are not full agreed remuneration.
Obs. 3 Expert-Graph alone	36	B	plain RAG/RAG+ExpertCoT C vs. LEGO CoT B	P: graph supplies the negative rule that no remuneration claim exists.
Obs. 4 synergy	7	D	single components A/B vs. LEGO Full D	P+F: the good-faith, no-present-benefit exception controls the result.
Obs. 5 strongest baseline	16	C	G-Retriever+ExpertCoT D vs. LEGO Full C	C: the private return claim is limited to principal plus legal fruits.

Table 4: **Six representative P/F/C case studies.**

D. It depends on whether the approval authority later ratifies the transfer.

Difference analysis. The legal core is consent for a combined transfer of contractual rights and obligations. Civil Code Articles 551, 555, and 556 require consent by the other party; Article 140 permits implied expression but does not make silence equal consent by default. RAG pipelines improve because retrieval supplies this major premise. LEGO Full adds a relation check in the P stage, preventing later performance or business efficiency from being converted into statutory consent.

C.2 QID 4: Stamped Blank Contract and Apparent Agency

Case. Zhang was a salesperson of an enterprise and carried blank contract forms bearing the enterprise’s official seal for external contracting. Because Zhang accepted kickbacks, the enterprise formally removed him and circulated an internal email about the removal, but it did not recover the stamped blank contracts. On the day after leaving the enterprise, Zhang concealed his departure and used one of the stamped blank contracts to sign a purchase and sale agreement with a counterparty. The counterparty only checked the authenticity of the seal and did not ask Zhang to produce proof of employment. The enterprise’s internal rules state that former employees may not use company contract forms. The enterprise also copied Zhang’s removal notice to all long-term business partners, including the counterparty.

Question and options. Because Zhang had been removed and had no current authorization, and because he did not produce any updated authorization

document, how should the purchase and sale agreement be characterized?

- A. It was not formed.
- B. It is invalid because Zhang lacked agency authority, and the counterparty was clearly negligent in ignoring the enterprise’s formal notice and relying only on past impressions.
- C. It is voidable.
- D. It was formed and became effective.

Difference analysis. Article 172 provides that an unauthorized agency act is effective if the counterparty has reason to believe authority exists. The civil-law graph lists stamped blank contracts as a typical apparent-authority appearance. Correct pipelines succeed because the P stage treats apparent agency as a special rule that can override bare no-authority reasoning. Plain RAG and LEGO CoT fail by overweighting absence of current authority.

Doctrinal caveat. The rewritten facts include notice of Zhang’s removal to the counterparty. If that notice is treated as effectively received and understood, reasonable reliance may be weakened. The benchmark gold and source explanation preserve the original apparent-agency conclusion, so this case should be presented as LEGO Full matching the benchmark’s rule hierarchy, not as a universal conclusion for all notice variants.

C.3 QID 1: Mandate Termination and Loss Compensation

Case. Party A orally authorized Party B to purchase timber on A’s behalf, and the parties agreed that B would receive remuneration for the service.

Pipeline	Pred.	P-Provision	F-Fact	C-Conclusion
Qwen3-8B zero-shot	B	Does not anchor the answer in Articles 551, 555, and 556.	Overweights actual operation, silence, and board participation.	Treats commercial efficiency and silence as implied consent.
Naive RAG + CoT	A	Identifies consent as required for transfer of rights and obligations.	Makes the lack of Company B's consent decisive.	Finds the transfer illegal and selects A.
G-Retriever + Expert-CoT	D	Lists the transfer-consent rules.	Extracts the key facts correctly.	Shifts the answer from consent to later administrative ratification.
LEGO Full	A	Combines Articles 551, 555, and 556 as validity conditions.	Treats silence and performance as insufficient substitutes for consent.	Concludes that the missing consent makes the transfer illegal.

Table 5: P/F/C comparison for QID 19.

Pipeline	Pred.	P-Provision	F-Fact	C-Conclusion
Naive RAG + CoT	B	Stops at the general no-authority rule and misses apparent agency.	Overweights removal, no updated authorization, and lack of checking.	Infers invalidity directly from lack of authority.
Naive RAG + Expert-CoT	D	Treats Article 172 apparent agency as the special effectiveness rule.	Uses the stamped blank contract as an appearance attributable to the enterprise.	Finds apparent agency and selects D.
LEGO CoT	B	Mixes Articles 171, 172, and 504 without stable priority.	Treats failure to verify current employment as decisive fault.	Incorrectly classifies the agreement as invalid.
LEGO Full	D	Gives Article 172 priority over the general no-authority rule.	Focuses on the unrecovered stamped blank contract as apparent authority.	Finds the agreement formed and effective.

Table 6: P/F/C comparison for QID 4.

B spent time and effort on the matter, repeatedly visited timber markets, and contacted several suppliers. B had already selected specific timber specifications and reported quotations and stock information to A, who raised no objection. Later, A no longer wanted the timber and telephoned B to cancel the mandate without mentioning compensation. B objected.

Question and options. A orally mandated B to purchase timber and agreed to pay remuneration. A now cancels the mandate. Given that B has performed substantial work and completed the main entrusted affairs, which statements are correct?

- A has no right to unilaterally cancel the mandate; otherwise A must compensate B's losses.
- A may unilaterally cancel the mandate, but only in writing.
- A may unilaterally cancel the mandate, but must compensate B's losses.
- Because B has performed substantial work, A must still pay the agreed remuneration so that B receives consideration for the work product.

Difference analysis. Article 933 gives both parties a discretionary termination right and then allocates loss compensation. The improvement is not retrieval but C-stage legal-effect matching: "compensate loss" and "pay agreed remuneration" are distinct consequences. ExpertCoT succeeds because it checks the option predicate against the exact statutory consequence.

C.4 QID 36: Emergency Rescue and Negotiorum Gestio

Case. Zhang was travelling in a scenic area when he saw a woman standing alone near a cliff with an unusual expression. When the woman jumped, Zhang urgently grabbed her clothing and pulled her back. During the rescue, Zhang's camera was damaged, his arm was scratched, and he advanced medical, bandaging, food, and lodging expenses. The woman's family later expressed gratitude and said they were willing to compensate lost work time. Zhang was a medical professional who encountered the incident on his way home from work.

Question and options. Given that Zhang spent time and money on the rescue, may Zhang ask the woman to pay a certain remuneration?

- He may request full remuneration.

Pipeline	Pred.	P-Provision	F-Fact	C-Conclusion
Naive RAG + CoT	CD	Finds Article 933 on discretionary termination and loss compensation.	Correctly identifies paid mandate and work already done.	Expands loss compensation into full agreed remuneration.
G-Retriever + CoT	CD	Identifies the mandate termination rule.	Uses work-performance facts too strongly to support D.	Overselects D and broadens the legal effect.
Naive RAG + Expert-CoT	C	Separates the termination right from the compensation duty.	Uses the work facts only to support loss, not full remuneration.	Selects only C.
LEGO Full	C	Keeps the paid-mandate loss rule clear.	Does not convert substantial work into a full remuneration claim.	Matches the option predicate and rejects D.

Table 7: P/F/C comparison for QID 1.

- B. He may not request remuneration.
- C. Based on fairness, he may request appropriate remuneration.
- D. He may request remuneration from the woman’s family.

Difference analysis. Article 979 allows reimbursement of necessary expenses and appropriate compensation for losses; it does not create remuneration. The civil-law graph explicitly states that the manager has no remuneration claim. LEGO CoT already succeeds because ExpertGraph provides this negative legal effect. Plain RAG and RAG+ExpertCoT fail because they see expense or compensation facts but miss the legal boundary of remuneration.

C.5 QID 7: Misdelivered Milk and Good-Faith Recipient

Case. Because of a delivery worker’s negligence, milk ordered by Wang was mistakenly placed in the milk box of Wang’s neighbor Zhang. Zhang did not understand why the milk was there and, without checking, directly took it out and discarded it. The milk package bore Wang’s name, address, and order number. Wang discovered the missing milk the next day and demanded compensation from Zhang. The milk was indeed Wang’s lawful property.

Question and options. After Zhang discarded another person’s milk that had been placed in his milk box, and given that Wang did suffer property loss and Zhang did discard the milk, how should Zhang’s conduct be characterized?

- A. It constitutes unjust enrichment because Zhang obtained a benefit without legal basis and caused loss.

- B. It constitutes tort liability, either as intentional destruction of another’s property or as negligent disposal causing loss.
- C. It constitutes unauthorized agency.
- D. It involves no legal wrong.

Difference analysis. The gold explanation treats Zhang’s lack of awareness as decisive. Zhang was a good-faith recipient; the milk no longer existed; no present benefit remained. The graph lists unjust-enrichment elements and the good-faith recipient’s present-benefit limitation. This case demonstrates synergy: ExpertCoT alone is pulled to A by the general unjust-enrichment rule, while graph-alone/free CoT drifts to B by over-reading negligence. LEGO Full combines the exception and the fact filter.

C.6 QID 16: Bank Overpayment and Business Profit

Case. When Party A withdrew money from a bank, a bank employee mistakenly overpaid A by 10,000 yuan because of a counting error. A knew the extra money was not owed, but used the 10,000 yuan as capital for business, quickly bought goods and resold them, and made a profit of 5,000 yuan, of which 2,000 yuan was cost for labor and management. After discovering a cash shortage, the bank traced the flow of funds and confirmed that the profit was directly generated from the overpaid money. One month later the bank discovered the overpayment and asked A to return the principal and corresponding gains. A refused.

Question and options. Given that A used the bank’s funds for profit and that the bank suffered an interest loss, which statement is correct?

- A. A does not need to return anything because the problem was entirely caused by the bank’s own mistake.

Pipeline	Pred.	P-Provision	F-Fact	C-Conclusion
Naive RAG + CoT	C	Conflates necessary expenses, appropriate compensation, and remuneration.	Overweights advanced expenses, lost work time, and family gratitude.	Uses fairness to create a remuneration claim.
Naive RAG + Expert-CoT	C	Uses P/F/C but misses the no-remuneration legal effect.	Treats compensation facts as a remuneration basis.	Selects C.
LEGO CoT	B	Graph expansion supplies the negative rule: no remuneration claim.	Treats family gratitude as neither mandate nor public reward.	Because the question asks about remuneration, selects B.
LEGO Full	B	Separates rescue immunity, expenses, compensation, and remuneration.	Expense facts do not change the legal nature of remuneration.	Rejects C and D, selects B.

Table 8: P/F/C comparison for QID 36.

Pipeline	Pred.	P-Provision	F-Fact	C-Conclusion
Naive RAG + Expert-CoT	A	Uses only the general unjust-enrichment elements and misses the good-faith limitation.	Emphasizes name, address, and Wang’s loss while weakening Zhang’s good faith.	Incorrectly finds unjust enrichment.
G-Retriever + Expert-CoT	B	Shifts to tort fault without stabilizing the good-faith recipient rule.	Treats failure to check as sufficient legal fault.	Incorrectly finds tort liability.
LEGO CoT	B	Rejects unjust enrichment but drifts to tort.	Still treats non-checking as enough fault.	Selects B.
LEGO Full	D	Checks both the general unjust-enrichment rule and the good-faith exception.	Centers good faith, no retained benefit, and no intentional destruction.	Rejects A and B, selects D.

Table 9: P/F/C comparison for QID 7.

- B. A should return the overpaid 10,000 yuan, with no other liability.
- C. A should return the overpaid 10,000 yuan and one month’s interest.
- D. A should return the overpaid 10,000 yuan, one month’s interest, and the entire 3,000 yuan net profit generated by using the money.

Difference analysis. The gold explanation gives the bank principal plus one month’s interest. The interest is the legal fruit of money; downstream business profit is not awarded to the bank under the original exam’s private return claim. Wrong traces have the right broad domain, unjust enrichment, but over-extend the legal consequence. LEGO Full succeeds by checking whether option D’s consequence is strictly authorized by the controlling rule and the benchmark explanation.

Doctrinal caveat. The civil-law graph contains a broad note that returnable benefit may include gains obtained by using the original object. Some modern unjust-enrichment analyses may therefore support a broader answer. The benchmark gold follows the original exam explanation; this case should be used to illustrate legal-effect boundary control, not a universal profit-return rule.

C.7 Cross-Case Lessons

Across the six cases, the observed failures are mostly P/F/C mismatches rather than answer-format errors. Retrieval helps when the missing major premise is the bottleneck (QID 19). Expert-CoT helps when a retrieved rule must be mapped to the exact option predicate (QID 1). ExpertGraph helps when the missing information is a negative legal effect or exception (QID 36). LEGO Full is strongest when both are needed: the graph supplies the right rule neighborhood, and P/F/C prevents the C step from expanding one legal consequence into another (QID 7 and QID 16).

D Dataset Card

LawExamQA_Civil

Pipeline	Pred.	P-Provision	F-Fact	C-Conclusion
G-Retriever + Expert-CoT	D	Uses an overbroad rule requiring return of all gains from the object.	Overweights that the business profit was produced by the overpaid money.	Awards the business profit to the bank.
LEGO CoT	D	Expands damages under Article 987 into full return of net profit.	Underweights labor, management, and business intervention.	Selects D.
Naive RAG + Expert-CoT	C	Limits the return scope to principal and legal fruits.	Distinguishes the bank’s use-of-money loss from business profit.	Selects C.
LEGO Full	C	Uses Articles 122 and 985 for return, without expanding Article 987 to D.	Treats principal and interest as the private return scope.	Rejects D, selects C.

Table 10: P/F/C comparison for QID 16.

Statistic	Value
Total items	723
<i>Question type distribution</i>	
Single-select	365
Multi-select	254
True/False	104
Subjective (converted to options)	26
<i>Hop count distribution</i>	
1-hop	342
2-hop	215
3-hop	104
≥4-hop	62
<i>Subject area distribution</i>	
Contracts	291
Property	167
Torts	84
Marriage & Family	48
Succession	25
Other	108
Avg. tokens	158.06
Avg. option tokens	14.82
Vocabulary Hit Rate (median)	0.095

Table 11: **LawExamQA_Civil Dataset Card.** Detailed statistics for the 723-item release; the 26 subjective questions converted to options form an overlapping subset.

We construct LawExamQA_Civil from publicly available Chinese National Judicial Examination questions in the civil-law domain. The dataset contains both objective multiple-choice items and subjective questions. To enable unified automatic evaluation, subjective questions are converted into option-style items while preserving their original legal-reasoning requirements.

For each question, we use the accompanying official explanation to identify the statutory provisions involved in the reasoning chain. The number of distinct provisions cited or required by the explanation is treated as the *hop count*, providing a direct measure of statutory reasoning complexity.

LexRAG_Civil

Statistic	Value
Total items	140
Avg. question chars	21.0

Table 12: **LexRAG_Civil Dataset Card.** Statistics for the LexRAG_Civil subset used in our cross-benchmark evaluation.

LexRAG (Li et al., 2025b) is the first benchmark targeting RAG systems in multi-turn legal consultation, containing 1,013 expert-annotated five-round dialogues paired with a 17,228-article candidate pool; each generated response is scored by an LLM-as-judge along five rubrics—*factuality*, *user satisfaction*, *clarity*, *logical coherence*, and *completeness*—with an expert-response anchor of 8/10. LexRAG-Civil is the civil-law subset obtained by filtering LexRAG for civil-law items and randomly sampling.

PLawBench_Civil

Statistic	Value
Total items	114
Avg. question chars	67.4

Table 13: **PLawBench_Civil Dataset Card.** Statistics for the PLawBench_Civil subset used in our cross-benchmark evaluation.

PLawBench (Shi et al., 2026) is a rubric-based benchmark with ~850 expert-curated questions across 13 practical legal scenarios, spanning three task families (public legal consultation, practical case analysis, legal document generation) and grounded in ~12,500 fine-grained expert rubric items. We use its *Practical Case Analysis* section, where each item is structured into four components—*Fact*, *Law/Provision*, *Reasoning*, and *Conclusion*—and each component is scored against dedicated rubrics; PLawBench-Civil is the

civil-law subset obtained by filtering and randomly sampling.

Judge models and scoring protocol. For LexRAG, we used Qwen3-30B-A3B as the evaluator and followed the original LexRAG five-dimensional LLM-as-a-judge rubric and scoring prompt. The same evaluator configuration and prompt were applied to all compared systems. For PLawBench, we used Gemini-3.0-Pro-Preview as the evaluator and followed the scoring protocol specified in the original PLawBench paper. We did not conduct a human-agreement calibration of either judge, the reported margins on these two benchmarks should therefore be read as evidence that LEGO is competitive with the strongest RAG baselines under each benchmark’s native protocol, not as a calibrated measure of superiority.

E LegalExpert-Domain Graph Construction

Expert-annotated legal graph

Expert annotators were compensated at fair market rates. They were licensed legal professionals who had passed the national judicial examination, graduated from reputable law schools, and were paid at the standard hourly rate for legally trained research assistants at the authors’ institution.

The following protocol reproduces the complete instructions provided to annotators. Annotators were instructed to identify the applicable Civil Code provisions and doctrinal concepts; annotate constitutive elements, factual conditions, exceptions, defenses, and legal effects; connect them using the predefined relation schema; consult judicial interpretations and representative cases only when statutory provisions were underspecified; use mainstream doctrinal views as the default while recording disputed alternatives in notes; and resolve disagreements through expert discussion. No task-related risks beyond those ordinarily associated with scholarly legal annotation were anticipated.

1. **Node extraction.** Chinese Civil Code parsed into article nodes with metadata (Part, Chapter, Section, Article number, full text); legal-concept nodes were extracted from a curated civil-law taxonomy and linked to the relevant articles.
2. **Edge annotation.** Our expert-curated legal graph organizes all civil code concepts in a hierarchical structure (Part → Chapter → Section

→ Article). Each node is annotated with its definition, constitutive elements, comparisons with related provisions, and temporally ordered legal conditions and consequences. Beyond structural organization, we further label legally operative relations among concepts, including order of application, rule priority, hierarchical dependencies, prerequisites, statutory exceptions, and other doctrinal links, each accompanied by a brief legal justification.

Civil Law ExpertGraph Construction

To construct the Civil Law ExpertGraph, we follow a hierarchical legal interpretation process inspired by source validity and doctrinal legal reasoning. The goal is not to annotate benchmark-specific answers, but to build a general-purpose normative graph that captures the structure of PRC civil law. The graph encodes legal concepts, statutory provisions, constitutive elements, exceptions, priority rules, condition–consequence relations, and legal effects. These annotations are constructed independently of LawExamQA_Civil questions, answer labels, and official rationales.

Hierarchical Legal Interpretation Framework

Civil-law knowledge is organized according to the validity and interpretive function of legal sources. We use the following source hierarchy:

Civil Code Articles → Judicial Interpretations → Guiding / Reference Cases → Civil-law Textbooks and Academic Doctrines.

Civil Code articles provide the primary statutory basis. Judicial interpretations clarify the scope and application of statutory provisions. Guiding and reference cases illustrate how abstract norms operate in concrete factual settings. Civil-law textbooks and academic doctrines are used to refine conceptual boundaries, resolve doctrinal ambiguities, and standardize expert annotations.

Correspondingly, different interpretive methods are applied at different stages. Literal interpretation is used to extract concepts and elements directly from statutory text. Systematic interpretation is used to locate each provision within the structure of the Civil Code and to identify relations among provisions. Purposive and sociological interpretation are used to clarify how rules function in practice. Doctrinal and comparative interpretation are used to refine disputed or abstract civil-law concepts.

Annotation Process

Stage One: Article-level extraction through

literal interpretation.

Annotators first examine Civil Code provisions and extract basic legal units from the statutory text. These units include legal concepts, subjects, objects, constitutive elements, legal acts, conditions, exceptions, and legal effects. For example, a provision on contract validity may be decomposed into nodes such as legal act, expression of intent, capacity, legality, validity, invalidity, and revocability. At this stage, the annotation remains close to the statutory language and avoids adding case-specific reasoning.

Stage Two: Systematic interpretation within the Civil Code.

Annotators then situate each provision within the broader structure of the Civil Code, including Book, Chapter, Section, and neighboring provisions. This stage identifies inter-provision relations such as general–special rule, prerequisite, limitation, exception, priority, parallel application, and condition–consequence relation. For example, rules on contract formation, contract validity, real-right transfer, liability allocation, and remedies are linked according to their doctrinal order of application rather than surface textual similarity.

Stage Three: Refinement using judicial interpretations and representative cases.

Where statutory provisions are abstract or underspecified, annotators consult judicial interpretations and representative cases to clarify the practical scope of legal elements. This stage is used to refine borderline concepts, exceptions, defenses, and legal consequences. For example, judicial interpretations may clarify when a third party is protected, when a contract-related remedy is available, or when a general civil-law rule is displaced by a more specific rule.

Stage Four: Doctrinal consolidation through civil-law theory.

Finally, annotators consult civil-law textbooks and academic doctrines to standardize concept boundaries and relation types. This stage is especially important for distinguishing doctrinally close concepts, such as validity versus effectiveness, contract obligation versus real-right transfer, termination versus rescission, liability for breach versus tort liability, and invalidity versus revocability. When multiple doctrinal views exist, annotators record the mainstream view as the default graph relation and preserve disputed views as notes or alternative annotations.

Quality Control

To ensure consistency, all annotations follow a predefined schema of node types and edge types. Node types include legal concept, statutory provision, constitutive element, factual condition, exception, defense, and legal effect. Edge types include hierarchy, prerequisite, specification, exception, priority, parallel application, condition–consequence, and remedy relation. Disagreements are resolved through expert discussion, with emphasis on whether the annotated relation reflects a general civil-law structure rather than a benchmark-specific answer path.

Independence from Evaluation Data

The Civil Law ExpertGraph is constructed independently from benchmark questions, answer labels, options, and official explanations. Public civil-law examination questions are used only for evaluating whether the graph-enhanced retrieval and reasoning framework improves multi-provision legal reasoning. They are not used as annotation sources for graph nodes or edges. Thus, the graph provides general normative signals for retrieval and reasoning, rather than item-level supervision.

Node types and error-oriented semantic links.

Our Civil Law Legal Knowledge Graph encodes expert legal knowledge through typed nodes and semantic links. These links are represented by parent–child structures, sibling structures, and explicit semantic annotations such as prerequisite, distinction, exception, priority, and legal effect. The node design is guided by the error patterns observed in our analysis.

In this design, *Concept* nodes guide the model to the correct legal institution; *Element* nodes force prerequisite checking; *Distinction* nodes prevent confusion between similar doctrines; *Exception* nodes expose provisos and defenses; *Effect* nodes constrain the interpretation of legal consequences; and *priority* nodes impose the proper order of legal reasoning.

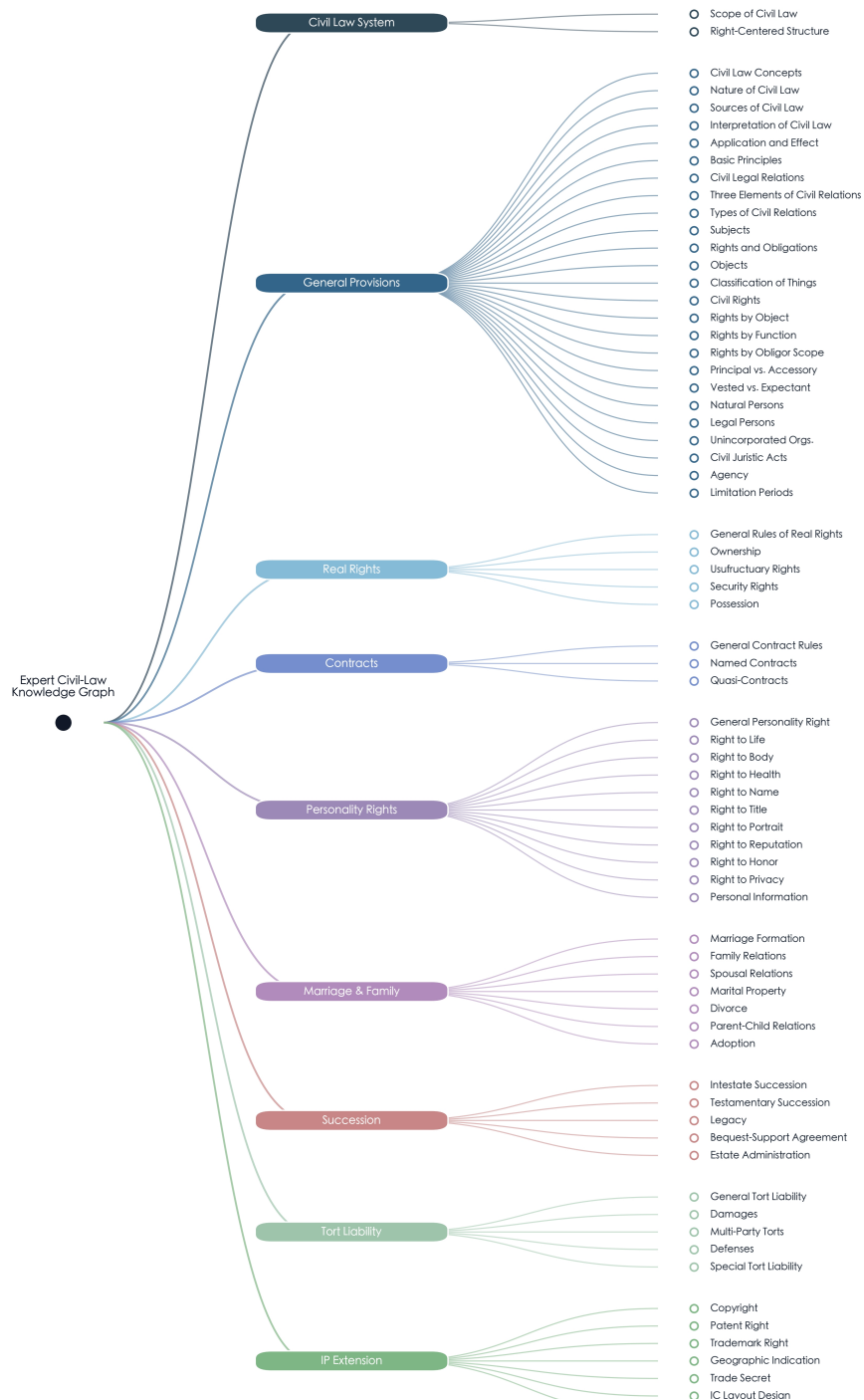


Figure 4: Legal Concept Tree of Civillaw KnowledgeGraph

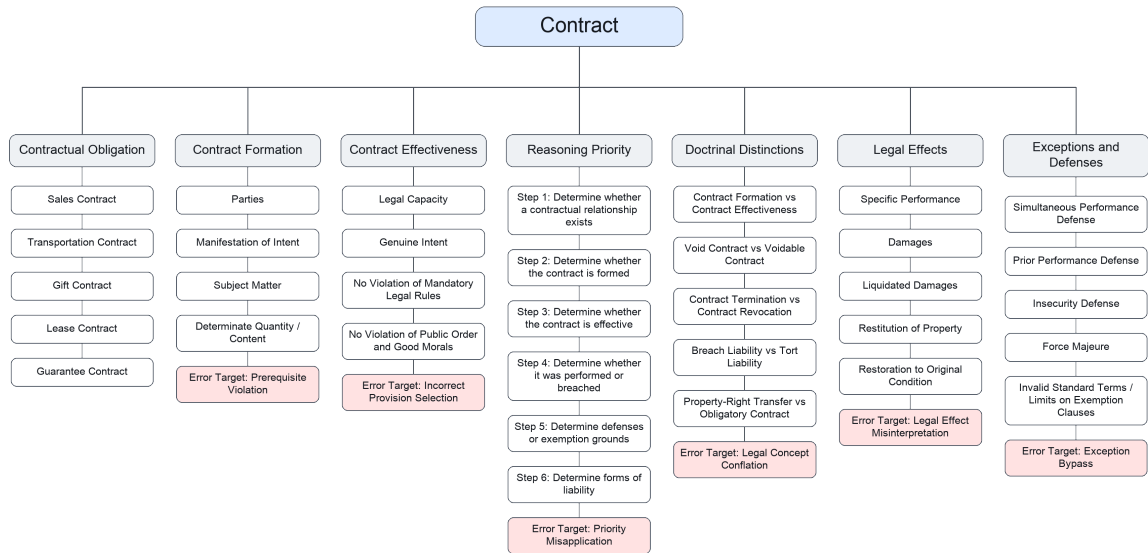


Figure 5: Legal relation in legal concept

Level	H1	H2	H3	H4	H5	H6	H7	H8	H9
Count	1	10	44	162	558	1,490	2,682	2,155	1,109

Table 15: Heading-depth distribution of the ExpertGraph source corpus.

F Civil Law ExpertGraph: Source Statistics

This appendix describes the structured Markdown corpus from which the Civil Law ExpertGraph is constructed. These figures characterise the upstream source and are not directly comparable with the deployed-graph statistics, because the downstream construction pipeline changes the unit of analysis through rule-unit typing, vocabulary normalisation, and article linking.

Overall. The source is a single hierarchical Markdown document of roughly 391K characters over 17,190 lines (8,717 non-empty), organised as a heading tree of nine depth levels. Its 8,211 heading nodes—one document title, ten book- or major-section headings, and 8,200 chapter-level and deeper nodes—form the candidate knowledge units consumed by the construction pipeline. The corpus resolves to 217 distinct Civil Code articles spanning the full range of the code (articles 1–1260).

Heading-tree depth. Heading depth is the primary structural cue. As Table 15 shows, the mass of the tree sits at H6–H9, where the corpus records legal-effect statements, exception clauses, and prerequisite enumerations—the material the pipeline maps onto normative edges.

Major-section breakdown. Table 16 reports heading counts for the ten top-level sections: a preface on the civil-law system, the seven books of the Civil Code, a supplementary chapter on intellectual property, and a notes section. Contracts, General Provisions, and Property Rights together account for roughly 71% of all hierarchical heading nodes, reflecting the doctrinal density of these three books.

Priority annotations. The corpus carries 357 in-text Priority: N annotations marking doctrinally salient nodes (208 at Priority 1, 149 at Priority 2). These are consumed downstream as weighting signals for retrieval boosting, graph-expansion priority during multi-hop traversal, rule-unit ranking when assembling the major-premise context, and legal-concept importance weighting.

Section	Headings (H3+)
Civil Law System	44
Book I General Provisions	1,912
Book II Property Rights	1,615
Book III Contracts	2,362
Book IV Personality Rights	376
Book V Marriage and Family	490
Book VI Succession	410
Book VII Tort Liability	607
Ch. 17 Intellectual Property	378
Notes	6
Total	8,200

Table 16: Per-section structural footprint of the source corpus.

Term	Freq.	Term	Freq.
Contract	1,514	Formation	261
Creditor’s right	713	Damages	260
Succession	708	Rescission	204
Property right	568	Termination	202
Agency	502	Void	196
Tort	385	Take effect	180
Security	365	Extinction	174
Mortgage	359	Constitutive element	143
Validity	351	Limitation period	121
Transfer	276	Claim right	113
		Lien	106
		Defence	83

Table 17: Top-22 legal concept frequencies. Counts are over the original Chinese terms; English glosses are given for readability.

High-frequency legal concepts. Table 17 reports frequencies for 22 core civil-law concepts. The distribution makes the doctrinal focus explicit: the corpus is anchored on contract, creditor’s right, succession, property right, agency, and tort, with substantial coverage of the machinery connecting them—validity, transfer, formation, damages, rescission, termination, and the validity-modality vocabulary. The presence of procedural and defensive vocabulary (limitation period, claim right, lien, defence) indicates that the source captures not only first-order legal concepts but the operational apparatus used in reasoning.

From source to deployed graph. The corpus above is the structured input to the construction pipeline. The deployed ExpertGraph is produced by typing each candidate heading node (concept / element / effect / exception-defence / relation / article), normalising vocabulary across hierarchical levels, resolving article references against the Civil Code text, and adding the normative edges—general/special, principle/exception, prerequisite/consequence, and related types—traversed by Syllogistic CoT. The figures in this appendix therefore characterise the corpus from which the graph is built, not the graph itself.

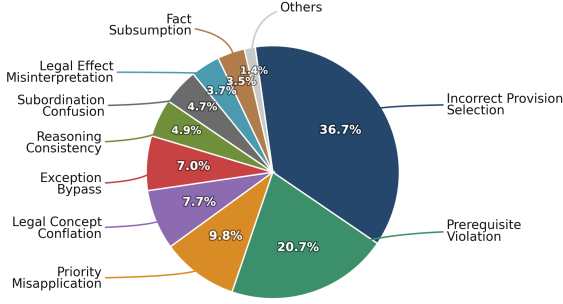


Figure 8: **Error-type distribution** on items mis-answered by full LEGO. These seven failure modes motivate the Syllogistic-CoT design choices in Section 3.2.

G Detailed Error Analysis

Table 18 reports the primary error labels for the 430 items mis-answered by the full LEGO setting. We code each item by the earliest reasoning step that makes the final answer unrecoverable. This convention is important because legal reasoning errors often cascade: a wrong provision may later produce an apparent priority conflict, or a missed prerequisite may later look like an erroneous legal effect. The analysis below therefore focuses on root causes rather than surface symptoms.

Figure 8 summarises the error-type distribution. Most failures occur before final answer selection: incorrect provision selection (36.74%) and prerequisite violation (20.70%) together account for 57.44% of all errors. The top seven categories account for 91.39%, indicating that wrong answers are concentrated in a small set of structural reasoning failures rather than diffuse factual uncertainty.

Incorrect Provision Selection. This is the largest category, covering 158 errors (36.74%). These cases usually arise when the model anchors the fact pattern to provisions that are lexically or topically similar but legally inapplicable. For example, a question involving carrier liability may trigger a nearby contractual damages provision while omitting the special transportation rule that controls the case. This error shows that retrieval is not merely a semantic matching problem: the selected provision must occupy the correct position in the legal topology and must be licensed by the facts.

Prerequisite Violation. Prerequisite violations account for 89 errors (20.70%). In these cases, the model applies a downstream legal consequence before verifying an upstream legal state, such as dis-

cussing breach liability before establishing contract validity, or assigning tort liability before checking the required causal relation. These errors directly motivate explicit prerequisite gates in Syllogistic-CoT: before deriving a conclusion Q , the model must verify every necessary premise P_i encoded by prerequisite edges.

Priority Misapplication. Priority misapplication appears in 42 errors (9.77%). The model often cites both a general rule and a special rule but applies them in the wrong order, or treats them as parallel rather than hierarchical. Such errors are especially common when civil law provisions contain nested exceptions, special chapters, or mandatory rules that override default contractual autonomy. This category motivates an explicit priority-resolution step before answer commitment.

Legal Concept Conflation. Legal concept conflation covers 33 errors (7.67%). These errors occur when the model merges formally distinct concepts, such as prerequisite versus trigger, presumption versus legal fiction, or solidarity versus supplementary liability. Although the generated explanation may sound legally plausible, the logical operator is wrong, which changes the validity of the inference. This supports our use of the 22-concept taxonomy as typed reasoning operations rather than loose textual labels.

Exception Bypass. Exception bypass accounts for 30 errors (6.98%). Here the model correctly identifies a general rule but fails to test whether an exception defeats it. This is a non-monotonic reasoning failure: adding an exception-bearing fact can reverse a conclusion that would otherwise follow. This motivates the exception-checking micro-constraint in ExpertCoT (§3.2), which requires the model to test the retrieved provisos and defences before accepting any conclusion derived from a general rule.

Reasoning Consistency. Reasoning consistency errors cover 21 cases (4.88%). These include contradictions between intermediate conclusions and the final answer, switching parties mid-chain, or using one interpretation in the explanation and another in option selection. These errors motivate a final trace-consistency check that verifies whether the selected option is actually entailed by the stated premises.

Error Type	Count	%
Incorrect Provision Selection	158	36.74
Prerequisite Violation	89	20.70
Priority Misapplication	42	9.77
Legal Concept Conflation	33	7.67
Exception Bypass	30	6.98
Reasoning Consistency	21	4.88
Subordination Confusion	20	4.65
Legal Effect Misinterpretation	16	3.72
Fact Subsumption	15	3.49
Others	6	1.40

Table 18: **Error-type distribution** on items mis-answered by full LEGO. These seven failure modes motivate the Syllogistic-CoT design choices in Section 3.2.

Subordination Confusion. Subordination confusion appears in 20 errors (4.65%). The model mistakes dependent legal relations for independent ones, such as treating accessory obligations, derivative rights, or supplementary liabilities as if they could exist without their principal legal relation. This category shows why the graph must encode structural dependence, not only topical relatedness among provisions.

Legal Effect Misinterpretation and Fact Subsumption. The remaining substantive categories are legal effect misinterpretation (16 errors, 3.72%) and fact subsumption (15 errors, 3.49%). The former involves deriving the wrong consequence from an otherwise correct rule, while the latter involves mapping facts to the wrong constitutive elements. These are smaller but still important categories because they occur after provision retrieval succeeds, indicating that closed-set legal reasoning still requires careful rule application.

Design Implications. The distribution suggests that Syllogistic-CoT should be organized around four safeguards. First, a provision-grounding step reduces incorrect provision selection by forcing the model to state the controlling rule as the major premise. Second, a prerequisite-verification step blocks downstream conclusions until all necessary legal states are established. Third, a priority-and-exception step handles special rules, mandatory rules, and defeaters before the final inference. Fourth, a consistency check aligns the intermediate legal trace with the selected answer. Together, these checks target the concentrated failure mass: the top seven categories cover 91.39% of all observed errors.

QID	Hop	Gold	LEGO-unique
627	1-hop	ABD	✓
716	2-hop	ABD	✓
544	3-hop	CD	✓
22	4+-hop	A	✓

Table 19: Four LawExamQA-Civil cases on which LEGO is correct and *all* 13 baselines (GPT-5, DeepSeek-V3, Qwen3-30B-A3B, GLM-4.7-Flash, Qwen3-8B, GLM-4-9B-chat, DISC-LawLLM, LegalOne, Naive RAG, HippoRAG 2, RAPTOR, G-Retriever, LightRAG) are wrong.

H Multi-Hop Case Studies on LawExamQA-Civil

This appendix complements the aggregate error-type taxonomy of Appendix G. There we coded LEGO’s 430 mis-answered items by failure mode; here we examine the opposite population—items on which LEGO answers correctly while every one of 13 strong baselines fails—to identify the specific reasoning patterns that drive LEGO’s hop-resilience.

Selection. We isolate the 20 items on which LEGO is correct while all 13 baselines are wrong, and select four cases spanning the four hop strata and four distinct reasoning failure patterns. For each case we report the case facts, the question, the answer distribution across the 14 systems, and the controlling provisions.

H.1 QID 627 (1-hop): Attachment to Immovable Property Overriding Value-Based Processing

Case. A picks up a jade stone in the mountains and asks his friend B to have it appraised. B, a provincial-level master craftsman with twenty years in jade carving, spends three months carving a *bagua taiji* design into the stone without A’s consent, using special techniques that substantially raise its value; an expert appraisal values the carved stone at RMB 800,000 and the design wins a municipal craft-innovation award. B later offers to pay A for the raw material. B’s father C, unaware of the provenance and without B’s consent, has workers mount the carved stone into the wall of his own house for good fortune. A dispute over the stone ensues.

Question. Given that B’s processing substantially increased the value of the stone, which statements about its ownership are *incorrect*?

Options.

- **A.** A acquires ownership.
- **B.** Because the processing added substantial value and B is a professional carver, B acquires ownership.
- **C.** C acquires ownership.
- **D.** Whoever acquires ownership must pay damages to the other two parties.

Gold. ABD. Civil Code Art. **322** (ownership of things produced by processing, attachment or mixture). The single article requires *two* separate determinations. (i) *Ownership*: the facts combine processing (B’s carving) with attachment to immovable property (mounting into C’s wall); under the “full use of the thing” principle the immovable absorbs the movable, so ownership vests in C. Statement C is therefore the only correct one, and A and B are incorrect — in particular, the value-based processing rule that would award the stone to the craftsman is displaced once attachment to immovable property occurs. (ii) *Indemnity*: Art. 322 distinguishes *damages*, owed by a party at fault, from *compensation*, owed by a faultless party. B, who carved another’s property without consent, is at fault; A and C are not. Statement D collapses this distinction and asserts an unconditional duty to indemnify both other parties, and is therefore also incorrect.

Answer		distribution.
Answer	Systems	
ABD	LEGO Full ✓	
AC	Qwen+IRAC, Naive RAG+IRAC, G-Retriever+IRAC, G-Retriever+Expert-CoT	
ABCD	Zero-shot, Naive RAG+Expert-CoT	
ABC	Naive RAG+CoT	
ACD	G-Retriever+CoT	
BCD	Qwen+CoT	
B	LEGO-CoT, LEGO-IRAC	

Table 20: Answer distribution on QID 627. Gold is ABD; only the full system recovers it.

Baseline failure mode. Baselines fail at one of the two determinations, and mostly at both. Those that retrieve Art. 322 read it through the value-based processing rule alone: having established that the carving added the dominant share of value, they accept B as correct and never test whether the subsequent attachment to C’s house displaces that rule. A second group misses the fault-sensitive indemnity structure and treats D — an unconditional duty to indemnify both parties — as an unobjectionable statement of fairness. Because the item

asks which statements are *incorrect*, an error on either determination flips the whole answer set.

LEGO’s contribution. LEGO resolves the two determinations in sequence: “the carving is processing, but the stone was subsequently attached to C’s house; under the principle of full use of the thing the immovable absorbs the movable, so ownership vests in C (rejecting A and B). Art. 322 further requires *damages* only from a party at fault — here B, who carved without consent — and *compensation* from a faultless acquirer, so D’s unconditional formulation is wrong.” The ExpertGraph edge linking the processing rule to the attachment rule inside Art. 322 supplies the priority that flat retrieval of the article text does not expose.

Pattern. SINGLE-ARTICLE, MULTI-DOCTRINE RESOLUTION: an item may cite one provision yet demand several independent determinations under it — here a displacement rule (attachment over processing) and a fault-sensitive remedy distinction (damages vs. compensation). Hop count measures the number of provisions in the rationale, not the number of determinations they require; this 1-hop item defeats all baselines precisely because its single article is internally structured.

H.2 QID 716 (2-hop): Threshold-Gated Mental-Distress Claim

Case. A and B are neighbours. A’s house leaks chronically, flooding the corridor B uses daily. B repeatedly demands repair; A refuses, and later publicly mocks B (“you can’t even walk past a puddle”). B develops anxiety symptoms (medical record on file); the village committee fronts RMB 1,000 to clear the water.

Question. Among the following claims, which are correct?

Options.

- **A.** B may demand removal of nuisance.
- **B.** B may demand elimination of the hazard.
- **C.** B may demand a public apology, because the prolonged flooding caused obvious mental pressure.
- **D.** B may demand reimbursement of the RMB 1,000.

Gold. ABD. Civil Code Arts. **236** (real-right protection: removal of nuisance / hazard elimination) and **980** (necessary expenses incurred by a third

party may be recovered). Option C requires the additional threshold of *serious* mental harm under Art. 1183; the facts do not meet that bar.

Answer		distribution.
Answer	Systems	
ABCD	GPT-5, DeepSeek-V3, Qwen3-30B-A3B, GLM-4.7-Flash, Qwen3-8B, DISC-LawLLM, Naive RAG, HippoRAG 2, RAPTOR, G-Retriever, LightRAG	
ABC	GLM-4-9B-chat	
AB	LegalOne	
ABD	LEGO	

Baseline failure mode. Eleven of thirteen baselines pick *everything*—ABCD—failing to test option C against the constitutive-element gate of Art. 1183. “Obvious mental pressure” (the wording in option C) is below the statute’s threshold of “serious mental harm”; granting an apology remedy without checking this element is a classical *exception bypass* (cf. Appendix G, error type “Exception Bypass”).

LEGO’s contribution. LEGO explicitly verifies the Art. 1183 threshold before accepting C, and rejects it: “B’s claim of prolonged mental pressure does not, on the facts, satisfy the ‘serious mental harm’ requirement of Art. 1183, so the apology remedy is not available even though the underlying tort is established.” Options A, B, and D each map cleanly onto a controlling provision (Art. 236 for A and B; Art. 980 for D), so LEGO accepts them.

Pattern. THRESHOLD-GATED REMEDY: when a remedy depends on satisfying a constitutive element (here, “serious” mental harm), explicitly check the element against the facts before granting the remedy. The Syllogistic-CoT major-premise expansion makes the threshold visible; baselines that skip the check end up over-claiming.

H.3 QID 544 (3-hop): Joint Tort vs. Separate Tort with Divisibility

Case. A and B are neighbours, each keeping one goat. The two goats are known to roam together. One morning both pens are left unlocked; the goats together eat C’s rare medicinal herbs bare. The damage scene shows interleaved hoofprints and mixed eating marks; the share consumed by each goat cannot be distinguished. A and B had a verbal agreement to take turns supervising, but neither honoured it on the day.

Question. Regarding A’s and B’s liability:

Options.

- **A.** A and B may each escape liability by proving they discharged the duty of care.
- **B.** Under Civil Code Art. 1168, A and B bear joint and several liability for the jointly-committed tort.
- **C.** If the quantity eaten by each goat can be determined, A and B bear corresponding several liability.
- **D.** If the quantity cannot be determined, A and B bear liability in equal shares.

Gold. CD. Civil Code Arts. 1171, 1172, 1245. The two goat-owners did not jointly commit a tort under Art. 1168; their tortious acts are *separate*, with damage that happens to combine (Art. 1172). When the divisible share can be proved, each bears proportional liability; when it cannot, the default is equal apportionment.

Answer		distribution.
Answer	Systems	
BCD	GPT-5, DeepSeek-V3, Qwen3-8B, LegalOne, Naive RAG, HippoRAG 2, RAPTOR, G-Retriever, LightRAG	
BD	Qwen3-30B-A3B, GLM-4-9B-chat	
BC	GLM-4.7-Flash	
D	DISC-LawLLM	
CD	LEGO	

Baseline failure mode. Nine of thirteen baselines pick BCD: they retain options C and D (the correct divisibility logic under Art. 1172) but *also* pick B (joint tort under Art. 1168). This is internally inconsistent—Art. 1168’s joint liability is incompatible with Art. 1172’s proportional liability—but baselines do not detect the conflict because they retrieve and apply each option’s apparent statute independently. The error reflects what Appendix G labels *priority misapplication*: when a general rule (joint tort) and a special rule (separate-act tort with divisibility) both surface in retrieval, baselines treat them as parallel rather than hierarchical.

LEGO’s contribution. LEGO’s response explicitly distinguishes three statutes: Art. 1168 (joint tort, requires concerted action), Art. 1171 (concurrent dangerous acts, each act alone sufficient to cause the entire damage), and Art. 1172 (separate-act tort, damage combines). It then applies the facts to Art. 1172 specifically (acts are separate; damage is combined and only conditionally divisible), and rejects Art. 1168 because the verbal supervision agreement was not actually executed—there is no

concerted action. The conclusion CD follows from Art. 1172’s two-branch structure (provable share → proportional; unprovable → equal).

Pattern. GENERAL-VS-SPECIAL SELECTION AMONG TORT REGIMES: ExpertGraph’s normative edges (joint vs. separate vs. concurrent-dangerous) gate which liability rule applies, preventing the silent merger of Art. 1168 with Art. 1172 that traps every baseline.

H.4 QID 22 (4+-hop): Independent Mortgage Transfer (Art. 406 vs. Art. 407)

Case. D holds a mortgage over C’s house. To secure a friend’s bank loan, D signs a written agreement transferring *the mortgage right alone* (not the underlying debt) to the bank. The transfer is duly registered; the registry lists the bank as the new mortgagee. The bank, relying on the registry’s public-faith principle, accepts the mortgage as security and disburses the loan.

Question. Given that the bank has completed registration and relied in good faith, is the mortgage-transfer contract unlawful?

Options.

- **A.** Yes.
- **B.** No — the mortgage is a freely transferable property right; registration completes the property-right transfer, and the bank has lawfully acquired the mortgage.
- **C.** No — effective upon notice to the debtor.
- **D.** No — effective upon C’s consent.

Gold. A. Civil Code Arts. 153, 216, 217, 407. Art. 407 is the controlling special rule: a mortgage right cannot be transferred independently of the underlying debt; the general transferability and registration-effect rules (Arts. 216, 217) are overridden in this configuration.

Answer **distribution.**

Answer **Systems**

B All 13 baselines: GPT-5, DeepSeek-V3, Qwen3-30B-A3B, GLM-4.7-Flash, Qwen3-8B, GLM-4-9B-chat, DISC-LawLLM, LegalOne, Naive RAG, HippoRAG 2, RAPTOR, G-Retriever, LightRAG

A **LEGO**

This is the most striking unanimous-baseline-failure case in the dataset: every closed-model baseline (including GPT-5 and DeepSeek-V3), every law-tuned LLM, and every RAG baseline picks B.

Baseline failure mode. Baselines apply the general rule (Arts. 216/217: registration completes real-property right transfer; public-faith protection for good-faith reliance on the registry) without checking whether a more specific rule constrains the rule’s domain. Art. 407, located in the mortgage chapter, explicitly removes *independent* mortgage transfers from the general transferability regime. The error is a textbook *priority misapplication*: the general rule is correctly identified but the special rule that overrides it is silently omitted.

LEGO’s contribution. LEGO’s response makes the rule hierarchy explicit: “Art. 406 is the general rule (transferability of the mortgaged property with the mortgage following); Art. 407 is the special rule (the mortgage right cannot be separated from the secured debt); the special rule prevails.” Applied to the facts (only the mortgage was transferred; the secured debt was not), the contract is unlawful and the registration cannot cure the substantive invalidity. Note that the public-faith principle (Art. 216) protects *procedural* reliance on the registry, not the substantive validity of an act prohibited by Art. 407—this nested distinction is what every baseline misses.

Pattern. GENERAL-VS-SPECIAL RULE RESOLUTION AT THE 4-HOP LEVEL: the case requires composing four provisions (153 invalidity, 216 public faith, 217 registration effect, 407 mortgage-debt inseparability) and resolving their priority. Without an explicit rule-hierarchy step—which Syllogistic-CoT enforces and ExpertGraph supports with normative edges—retrieval that surfaces only Arts. 216/217 yields a confident but wrong answer.

H.5 Cross-Case Synthesis

The four cases map onto four reasoning failure patterns that ExpertGraph and Syllogistic-CoT are specifically designed to address. Three of the four also appear among the top error types in Appendix G’s aggregate taxonomy (the analysis of items on which LEGO *fails*), confirming that LEGO’s wins and losses share a common skeleton: success requires explicit rule-hierarchy resolution and constitutive-element checking, while failure occurs when those steps are skipped or applied to the wrong rule.

Reasoning pattern	Case	Aggregate-taxonomy analogue
Right-holder disambiguation among topically related parties	QID 174	Legal Concept Conflation
Threshold-gated remedy (constitutive-element check before grant)	QID 716	Exception Bypass
General-vs-special selection among tort regimes	QID 544	Priority Misapplication
General-vs-special rule resolution at 4-hop	QID 22	Priority Misapplication / Incorrect Provision Selection

Two observations follow. First, the unanimous-baseline-failure pattern on QID 22 is not noise: Art. 407 is doctrinally required but lexically remote from the case facts, so similarity-driven retrieval does not surface it, whereas rule-card routing does. Second, the 4+-hop margin of +6.4 pp over the strongest LLM baseline is not driven by retrieving more provisions, but by retrieving the right *configuration* of provisions and resolving their hierarchy. The aggregate hop-resilience of LEGO (Table 1) is therefore a population-level signature of the per-case mechanisms documented in this appendix.

Configuration	1h	2h	3h	4+h	All
Qwen3-8B (zero-shot)	25.4	27.4	26.9	16.1	25.4
Qwen3-8B + CoT	28.9	27.0	31.7	33.9	29.2
Qwen3-8B + IRAC-CoT	30.1	34.4	26.0	33.9	31.1
Naive RAG + CoT	34.2	33.0	35.6	37.1	34.3
Naive RAG + IRAC-CoT	32.5	31.2	37.5	25.8	32.2
G-Retriever + CoT	34.2	34.0	30.8	35.5	33.7
G-Retriever + IRAC-CoT	33.3	34.9	27.9	38.7	33.5
Naive RAG + ExpertCoT	33.3	40.5	34.6	35.5	35.8
G-Retriever + ExpertCoT	34.8	38.1	32.7	35.5	35.5
ExpertGraph + CoT	36.5	33.5	37.5	33.9	35.5
ExpertGraph + IRAC-CoT	36.5	39.1	35.6	38.7	37.3
Full system (LEGO)	41.8	40.5	37.5	38.7	40.5

Table 21: Accuracy (%) on LawExamQA-Civil by hop count for twelve component configurations. Single-component upgrades give modest gains; combining ExpertGraph with ExpertCoT yields a super-additive gain.

I Component-Synergy Case Studies on LawExamQA-Civil

This appendix isolates the internal contribution of the two LEGO components — the ExpertGraph retrieval module and the ExpertCoT reasoning module — by examining a 5×3 ablation grid and the case-level behaviour at its extremes. It complements Appendix H (cases where LEGO beats external baselines) and Appendix G (aggregate error taxonomy on LEGO failures): together the three appendices document, respectively, what LEGO adds over baselines, what LEGO adds over its own ablations, and where LEGO still fails.

Setup. We instantiate twelve configurations crossing five retrieval modules (none, Naive RAG, G-Retriever, ExpertGraph, Full) with three reasoning modules (plain CoT, IRAC-CoT, ExpertCoT). Twelve of the fifteen cells were run; the missing three are not informative for our hypothesis.

Accuracy matrix. Table 21 reports per-configuration accuracy on each hop stratum. Three patterns are visible. First, swapping Naive RAG for ExpertGraph (with the same plain CoT) yields a small +1.2 pp gain on the aggregate but a +2.3 pp gain on 1-hop. Second, swapping plain CoT for ExpertCoT (with the same Naive RAG) yields a +1.5 pp gain on the aggregate but a +7.5 pp gain on 2-hop. Third, combining both yields +6.2 pp on the aggregate and +7.6 pp on 1-hop — substantially more than the sum of the two single-component gains.

Hop	Items	Full-unique-correct
1-hop	342	5
2-hop	215	6
3-hop	104	1
4+-hop	62	2
Total	723	14

Table 22: Items on which the Full LEGO system is correct and every one of the eleven ablations is incorrect.

Synergy decomposition. Taking Naive RAG + plain CoT as the baseline (34.30%), we isolate the contribution of each LEGO component and their interaction:

- ExpertGraph retrieval alone (replacing Naive RAG, keeping plain CoT): 35.55%, **+1.24 pp**.
- ExpertCoT alone (keeping Naive RAG, replacing plain CoT): 35.82%, **+1.52 pp**.
- Both together (Full): 40.53%, **+6.22 pp**.
- **Synergy** : $6.22 - (1.24 + 1.52) = +3.46 \text{ pp}$.

The synergy term is more than twice the magnitude of either single-component gain. Substantively, this means the components are not independent quality boosters: ExpertGraph supplies provisions whose joint application requires ExpertCoT to discriminate general from special rules, and ExpertCoT requires ExpertGraph’s normative edges to be supplied as input rather than re-discovered from flat similarity.

Case selection. We then identify items where the Full system answers correctly while *every one of the eleven ablations* (including the strong ExpertGraph + IRAC-CoT variant and the two ExpertGraph alternatives) is incorrect. Across the 723 items there are 14 such cases (Table 22). We expand three of them spanning the 2/3/4+-hop strata.

I.1 QID 237 (2-hop): Constitutum Possessorium

Case. A owns a jade worth RMB 10,000. A and B sign a sale contract for RMB 11,000, “effective upon signature”; B is to take delivery in three days. After signing, A asks to keep the jade for personal enjoyment a few days more; B agrees but no separate property-transfer document is signed. The jade remains in A’s physical custody. The next day A re-sells and delivers the jade to a knowing buyer C, and the chain continues to D, E (a finder after loss), F.

Question. Given that the sale contract is effective and B has confirmed ownership at signature, which statements about B’s acquisition of ownership and delivery are correct?

Options.

- **A.** B acquires ownership upon contract validity.
- **B.** B acquires ownership when the borrowing agreement takes effect.
- **C.** Because there is no actual delivery, ownership has not transferred.
- **D.** A delivered the jade to B by means of *constitutum possessorium*.

Gold. BD. Civil Code Arts. **224** (general rule: ownership of movables transfers upon delivery) and **228** (*constitutum possessorium*: when the parties agree the transferor continues to possess, ownership transfers at the moment the agreement takes effect). Art. 228 is the controlling *special* rule.

Ablation behaviour. The agreement to “borrow back” the jade after the sale is the textbook *constitutum possessorium* pattern that activates Art. 228; nine of eleven ablations select CD — correctly rejecting A but *also* rejecting B and accepting C. They thus apply the general delivery rule (Art. 224) and miss the special rule. Even the strongest ablation, ExpertGraph + IRAC-CoT, retrieves both Arts. 228 and 224 but still answers CD: retrieval is sufficient but reasoning fails to resolve their hierarchy. Only the Full system (ExpertGraph + ExpertCoT) explicitly invokes the general-vs-special priority and selects BD.

Per-configuration outcomes.

Configuration	Pred	OK
Qwen3-8B (zero-shot)	AD	✗
Qwen3-8B + CoT	CD	✗
Qwen3-8B + IRAC-CoT	CD	✗
Naive RAG + CoT	C	✗
Naive RAG + IRAC-CoT	CD	✗
G-Retriever + CoT	CD	✗
G-Retriever + IRAC-CoT	CD	✗
Naive RAG + ExpertCoT	CD	✗
G-Retriever + ExpertCoT	CD	✗
ExpertGraph + CoT	C	✗
ExpertGraph + IRAC-CoT	CD	✗
Full (LEGO)	BD	✓

Full-system trace (excerpt).

Art. 228 is the special rule and overrides Art. 224’s general delivery requirement. Since A and B agreed that A would continue to possess the jade after the sale, ownership transfers at the moment of the agreement (B), not upon physical

*delivery (rejecting C). The delivery itself takes the form of *constitutum possessorium* (D).*

What the ablation isolates. ExpertGraph’s normative edge between Arts. 224 and 228 is necessary but not sufficient: ExpertGraph + IRAC-CoT retrieves both articles but cannot apply the general-vs-special priority on its own. ExpertCoT, in turn, is necessary to consume that edge and produce the correct ordering. The two components are jointly responsible for the +12.2 pp gap on this single case.

1.2 QID 509 (3-hop): Title-Retention Sale plus Good-Faith Third Party

Case. Zhou sells a computer to Wu under a title-retention contract for RMB 6,000 in five monthly instalments of RMB 1,200; until full payment, title remains with Zhou. Wu pays the first four instalments on time but defaults on the fifth. The computer malfunctions; Wu hands it to Zhou for repair. After repair Zhou, citing Wu’s breach, re-sells the computer to Wang for RMB 6,200; Wang checks Zhou’s original purchase receipt and accepts in good faith.

Question. Based on the title-retention rules, which statements are correct?

Options.

- **A.** Wang may acquire ownership of the computer.
- **B.** Because title remains with Zhou, Zhou may exercise the right of recovery when Wu cannot pay the final instalment.
- **C.** If Wu’s unpaid due amount reaches RMB 1,800, Zhou may demand the full purchase price.
- **D.** If Wu’s unpaid due amount reaches RMB 1,800, Zhou may rescind the contract and demand a use fee for the computer.

Gold. ACD. Civil Code Arts. **311** (good-faith acquisition), **634** (one-fifth default-acceleration rule in instalment sales), **642** (seller’s right of recovery under title retention, conditional on notice and reasonable period). The case requires composing three rules: B is incorrect because the right of recovery requires *notice plus reasonable period* (Art. 642), not just inability to pay; A is correct via Art. 311 (Wang’s good-faith reliance on the receipt); C and D both follow from Art. 634 (1/5 threshold of RMB 1,200 is exceeded by RMB 1,800).

Ablation behaviour. The ablations fragment across the four options because each lacks one of the three doctrinal pieces:

- Six of eleven ablations select **AD** (good-faith acquisition + rescission), missing Art. 634’s full-price branch (C).
- Naive RAG + ExpertCoT selects **BCD**, retaining the wrong recovery-right framing (B).
- Two G-Retriever variants select **AB** (mixing good-faith acquisition with the unconditional recovery framing).

ExpertGraph + IRAC-CoT retrieves Arts. 642 and 634 but still answers AD; the Full system is the only configuration that retrieves the same articles *and* composes them into the three-rule chain (Art. 311 for A; Art. 634 for both C and D, distinguishing its two branches; Art. 642 as the gate that rejects B).

Per-configuration outcomes.

Configuration	Pred	OK
Qwen3-8B (zero-shot)	ABCD	✗
Qwen3-8B + CoT	A	✗
Qwen3-8B + IRAC-CoT	AD	✗
Naive RAG + CoT	AD	✗
Naive RAG + IRAC-CoT	AD	✗
G-Retriever + CoT	AD	✗
G-Retriever + IRAC-CoT	AB	✗
Naive RAG + ExpertCoT	BCD	✗
G-Retriever + ExpertCoT	AB	✗
ExpertGraph + CoT	AD	✗
ExpertGraph + IRAC-CoT	AD	✗
Full (LEGO)	ACD	✓

What the ablation isolates. This case demonstrates the limit of single-component upgrades. With ExpertGraph retrieval alone, the model has access to Art. 634 but does not unpack its two-branch structure (full-price acceleration vs. rescission-plus-use-fee); with ExpertCoT alone over Naive retrieval, the model develops the right reasoning skeleton but misses the controlling articles. The Full system both retrieves and decomposes, recovering all three of A, C, and D.

I.3 QID 652 (4+-hop): Will Validity vs. Mandatory Reservation for a Fetus

Case. Zhou and Wu, married for years and childless, signed an artificial-insemination consent at a hospital; Wu became pregnant. In April 2021, Zhou (now hospitalised with cancer) wrote and signed a handwritten holographic will: “All housing purchased after marriage is inherited by my parents.” The will is signed and dated, read aloud to the parents, and otherwise satisfies every form

requirement of Civil Code Art. 1134. Zhou died in May 2021; Wu gave birth in October 2021. Zhou’s parents immediately listed the house for sale.

Question. Given that the will reflects Zhou’s genuine intent and is formally complete, which of the following statements are *incorrect*?

Options.

- **A.** Because artificial insemination raises ethical issues, the AI consent is void as contrary to public order and good morals.
- **B.** Being a formally complete and genuinely intended holographic will, Zhou’s will is fully valid.
- **C.** Because Wu’s child was conceived by artificial insemination rather than naturally, the child is not a marital child of Zhou and Wu.
- **D.** When dividing Zhou’s estate, an inheritance share must be reserved for the fetus in Wu’s womb.

Gold. **ABC.** Civil Code Arts. **153** (juristic-act validity), **1071** (parent-child relationship covers AI children), **1153** (inheritance division), **1155** (*mandatory share reservation for a fetus*). Statement D is the only correct one; A, B, and C are each separately wrong. The non-obvious wrong is B: a will may be formally complete *and* substantively over-broad if it disposes of estate without reserving the share that Art. 1155 makes mandatory for a conceived but unborn heir.

Ablation behaviour. All eleven ablations select **AC**. They correctly reject A (AI consent is not contra public order) and C (AI children are marital children under Art. 1071) but accept B: they treat “formally complete” as a sufficient condition for “fully valid,” silently overlooking the mandatory-reservation requirement under Art. 1155. The error pattern is *exception bypass* (cf. Appendix G): even after retrieving the form-validity provisions, the ablations fail to test the will against the substantive limit imposed by Art. 1155 on every disposition of an estate when a heir is in gestation. The Full system uniquely retrieves Art. 1155 *and* reasons through to the substantive-invalidity conclusion.

Per-configuration outcomes.

Case	ExpertGraph supplies	ExpertCoT supplies	Joint effect
QID 237 (2-hop)	Both Arts. 224 and 228 in the retrieval set	General-vs-special priority resolution	Selects BD instead of CD
QID 509 (3-hop)	Arts. 311, 634, 642 in a single retrieval	Two-branch decomposition of Art. 634	Selects ACD instead of AD
QID 652 (4+-hop)	Art. 1155 alongside form-validity articles	Substantive-limit check on a formally valid act	Selects ABC instead of AC

Configuration	Pred	OK
Qwen3-8B (zero-shot)	AC	✗
Qwen3-8B + CoT	AC	✗
Qwen3-8B + IRAC-CoT	AC	✗
Naive RAG + CoT	AC	✗
Naive RAG + IRAC-CoT	ACD	✗
G-Retriever + CoT	AC	✗
G-Retriever + IRAC-CoT	AC	✗
Naive RAG + ExpertCoT	AC	✗
G-Retriever + ExpertCoT	AC	✗
ExpertGraph + CoT	AC	✗
ExpertGraph + IRAC-CoT	AC	✗
Full (LEGO)	ABC	✓

Full-system trace (excerpt).

Art. 1143 governs the formal-validity of a holographic will; Art. 1155 is the special rule that requires a reservation for an in-gestation heir. The two are not mutually exclusive: a will may pass formal validity (under Art. 1134/1143) but fail substantive validity because it disposes of the entire estate without reserving the mandatory fetus share. The disposition to Zhou’s parents leaves nothing for Wu’s unborn child, violating Art. 1155.

What the ablation isolates. Both components are necessary on this case. ExpertGraph alone (with plain CoT or IRAC-CoT) retrieves form-validity provisions and the parent-child article but does not surface Art. 1155 as a substantive limit on the will’s scope; ExpertCoT alone (with Naive or G-Retriever retrieval) supplies the constitutive-element checking framework but lacks the article. Only the Full system retrieves Art. 1155 *and* applies the element-check that connects “unborn fetus” to the reservation requirement.

I.4 Cross-Case Synthesis

The three ablation cases instantiate a common pattern: each component supplies a necessary but insufficient capability, and the Full system’s win arises from their interaction rather than from either alone.

Three more general observations follow. First, the strongest single-component ablation in the matrix — ExpertGraph + IRAC-CoT at 37.3% — is still 3.2 pp below the Full system, and on the three

cases above it produces exactly the same wrong answer as the much weaker ablations: retrieving the controlling article is not the same as applying it. Second, the synergy gain of +3.46 pp is concentrated on items requiring multi-statute composition (general-vs-special priority, multi-branch rule decomposition, substantive-limit checking), which is precisely the regime that Syllogistic-CoT is designed to handle and that flat retrieval cannot reach. Third, the alignment between this synergy mechanism and the aggregate error-type taxonomy of Appendix G is exact: the dominant failure categories on LEGO’s 430 remaining errors are *Incorrect Provision Selection* and *Prerequisite Violation*, the same two failure modes that each of the eleven ablations exhibits on the three cases above. LEGO is not immune to these failures, but the ablation matrix shows that both ExpertGraph and ExpertCoT measurably reduce them, and that the two together reduce them more than either alone.

ID	Label	LEGO	Best B.	Δ
civil-107	Family/Marriage	90.9	81.8	+9.1
civil-682	Legal-theory Appl.	50.7	30.7	+20.0
civil-56	Individual Life	78.6	92.9	-14.3
civil-633	Cross-border Affairs	80.0	58.3	+21.7

Table 23: Four PLawBench-Civil case studies. civil-56 is included as a partial win: LEGO ranks second among three methods that retrieve the controlling statute, contrasting with three baselines that retrieve none.

J PLawBench-Civil Case Studies

This appendix expands the PLawBench-Civil row of Table 2 into case-level analyses. PLawBench-Civil scores each response on four rubric components (LAW, FACT, REASONING, CONCLUSION) and a composite Overall score; component scores are rationals in $[0, 1]$ and the Overall is reported in $[0, 100]$. We select four samples from the 114-sample evaluation set on which LEGO obtains the largest Overall margin over every baseline.

Selection. We enumerate all 114 samples, rank by $\text{Overall}_{\text{LEGO}} - \max_b \text{Overall}_b$, and pick the four cases below to span the four largest dataset labels (*family/marriage*, *individual life*, *legal-theory application*, *cross-border affairs*) and four distinct baseline failure modes.

J.1 civil-107 (Family/Marriage): Joint Marital Debt under Long Separation

Question. *Do debts incurred by one spouse in their own name constitute joint marital debt?* A divorcing husband claims six personal-name debts totalling over RMB 1.7M (bank loans, supplier debts, family-friend loans, online lending) are joint marital debts because they were used for “household consumption, business turnover, rent, medical, child education.” The wife denies knowledge of every debt. The spouses had been separated since 2017.

Reference. *Not joint debt.* The required analysis integrates “debt formation time, nature, use, household-necessity test, joint-business test, joint intent, and separation status.” Controlling provisions: Civil Code Art. 1064 (three categories of joint debt) and Art. 1089 (post-divorce repayment).

Per-method scores.

Method	L	F	R	C	Overall
Naive RAG	0.50	1.00	1.00	0	81.8
HippoRAG 2	0.50	0.75	0.50	0	54.5
RAPTOR	0.50	0.75	0.75	1	72.7
G-Retriever	0.50	1.00	0.75	0	72.7
LightRAG	0.50	1.00	0.75	0	72.7
LEGO	0.50	1.00	1.00	1	90.9

Baseline failure mode. Four of five baselines retreat into a non-committal conditional conclusion (“partially joint, partially not” / “may constitute joint debt subject to further review”). The PLawBench judge marks all four as conclusion-incorrect because the reference requires a directional judgment (*not joint debt*) qualified by the Art. 1064 *but-clause*.

LEGO’s contribution. LEGO’s conclusion mirrors the reference’s logical form (negation + exception clause):

The debts incurred by Zhang in his own name do not constitute joint marital debt, except for those debts that can be proven to have been used for joint marital life, joint production and operation, or to be based on joint marital intent.

In the reasoning section, LEGO is the only method that explicitly assigns the burden of proof to the creditor and connects this to the long-separation fact, aligning with the reference’s *actori incumbit probatio* requirement under Art. 1064 §2. Naive RAG cites the correct Supplementary Interpretation Art. 35 but dilutes the same content into a conditional sentence, costing the conclusion point.

Takeaway. On bivalent legal judgments with a *but-clause*, baseline RAG models often hedge; LEGO commits to the directional answer while preserving the exception. This is the cleanest LEGO win in the PLawBench dataset.

J.2 civil-682 (Legal-theory Application): Lease Continuity after Partnership-to-LLC Conversion

Question. A general partnership (“Hangzhou Qiantang Metal Products Factory”) leases a factory from Z at RMB 100k/month. The three partners later convert the partnership into a limited-liability company (“Qiantang Metal Products Co., Ltd.”) with the same core trade name; on May 2 the LLC’s legal representative declares: “Qiantang Co., Ltd. ratifies all transactions previously signed by A and B.” The LLC pays rent on time but Z now wants to reclaim the factory. May Z reclaim it?

Reference. Z cannot reclaim. The LLC’s possession is *rightful possession*. Required provisions: Civil Code Arts. **235** (return claim against wrongful possession), **462** (possessor’s return right), **733** (lease return on expiry), **967** (definition of partnership contract), and **551** (debt transfer requires creditor consent).

Per-method scores.

Method	L	F	R	C	Overall
Naive RAG	0.00	0.75	0.00	0.40	22.7
HippoRAG 2	0.25	0.75	0.00	0.60	30.7
RAPTOR	0.00	0.70	0.00	0.40	21.3
G-Retriever	0.00	0.70	0.00	0.40	21.3
LightRAG	0.00	0.75	0.00	0.40	22.7
LEGO	0.25	1.00	0.33	0.60	50.7

Baseline failure mode. All six methods correctly conclude “Z cannot reclaim,” so conclusion polarity is shared. The discriminator is doctrinal framing. Five of six baselines collapse the argument onto Civil Code Art. 75 (acts performed during legal-person setup are inherited by the legal person). This anchors the case in a *setup-act* doctrine, but the reference requires the *partnership-contract conversion* doctrine. Failure-mode variants: LightRAG invokes Arts. 96/98 to argue the partnership and the LLC are distinct subjects, then must reverse itself to accept the ratification, producing an internally contradictory argument; RAPTOR retrieves Art. 547 (assignment of claims) and Art. 985 (unjust enrichment), both topical mismatches; G-Retriever retrieves Art. 141 (withdrawal of intent), unrelated.

LEGO’s contribution. LEGO is the only method that retrieves Art. 967—one of the five reference provisions—anchoring the analysis in the partnership-contract doctrine rather than the generic setup-act doctrine. It additionally cites Art. 75 (setup-act inheritance) and Art. 703 (lease form), producing a three-tier argument: *partnership contract* → *conversion* + *ratification* → *ongoing lease*. LEGO is also the only method whose case-facts section preserves the pivotal datum “Qiantang Co., Ltd. has paid rent on time,” achieving Fact = 1.00 where baselines score 0.70–0.75.

Takeaway. This case illustrates retrieval-precision benefits beyond conclusion correctness. LEGO finds a *small but pivotal* provision (Art. 967) that anchors the correct doctrinal frame. Baselines retrieve a generally relevant but doctrinally

off-target provision (Art. 75) and amplify its prominence at the cost of the controlling rule.

J.3 civil-56 (Individual Life): Self-help in a Neighbour Dispute

Question. Plaintiff smashed 2.5 m of his neighbour’s stone railing with a hammer, claiming the railing blocked ventilation and light; he was administratively detained for 8 days. Plaintiff argues the act was lawful self-help. Does the self-help defence stand?

Reference. Self-help does not stand. Controlling provision: Civil Code Art. **1177** (self-help requires *urgency, necessity, and immediate post-act notification of state authorities*).

Per-method retrieval and scores.

Method	Core articles cited	1177?	Overall
Naive RAG	1165 / 296 / Public Security 49	✗	50.0
HippoRAG 2	1177 + 1165 + 293	✓	78.6
RAPTOR	1165 / 288 / 296	✗	50.0
G-Retriever	233 / 235 / Criminal 275	✗	50.0
LightRAG	1177 + 1184	✓	92.9
LEGO	1177 + 238 + Public Security 49	✓	78.6

Baseline failure mode. All six methods reach the correct conclusion (self-help fails). The discriminator is whether Art. 1177—the *only* controlling statute—is retrieved. Three of five baselines miss it entirely and instead retrieve general-tort (Art. 1165), neighbour-relations (Arts. 288/296), property-protection (Arts. 233/235), or even Criminal Code Art. 275. These three baselines compensate by writing a generic “tort plus neighbour” argument that the judge marks as off-rubric, capping Overall at 50.

LEGO’s contribution. Once Art. 1177 is retrieved, LEGO is the only method besides the reference itself that *explicitly decomposes the article into three sub-elements*—urgency, necessity, immediate notification—and checks each against the facts:

Self-help requires three conditions: “urgent circumstances,” “inability to obtain timely protection from state authorities,” and “failure to act would cause irreparable damage.” ... The conduct occurred during daytime, and Y had already reported to the police ... so it does not satisfy “urgent circumstances” or “inability to obtain timely state-authority protection.”

This element-by-element structure mirrors the reference reasoning’s three subsections. Baselines that retrieve Art. 1177 (HippoRAG 2, LightRAG) achieve comparable or slightly higher Overall on

this single case via a more terse statute application; LEGO’s structured element-check, while not the absolute best on this individual case, is the form that the PLawBench rubric *generally* rewards across the rest of the benchmark.

Takeaway. When the controlling statute is reachable by single-shot retrieval, LEGO is one of three methods that find it. LEGO’s distinctive contribution is then the *element-by-element check* grounded in the major-premise structure of the syllogism (Section 3.2), not simply locating the article.

J.4 civil-633 (Cross-border Affairs): Validity of Cross-border Marriage Brokerage

Question. A Chinese plaintiff pays RMB 168,000 to an intermediary for a “Vietnamese bride” arrangement; the bride disappears one month after the marriage registration. The intermediary refunds only RMB 18,000. Is the brokerage valid? Must the intermediary refund the remainder?

Reference. The brokerage is *invalid* (violates Civil Code Art. **1042** ban on marriage-trade and the State Council 1994 Notice on cross-border marriage brokerage). The intermediary must refund partially, since the plaintiff bears co-fault as a legally competent adult. Controlling provisions: Civil Code Arts. **1042, 153, 157**.

Statute-retrieval pattern.

Method	Marriage-prohibition article	157?	Overall
Naive RAG	1048 ✗ (consanguinity, wrong topic)	✓	58.3
HippoRAG 2	1048 ✗	✓	50.0
RAPTOR	1048 ✗	✓	41.7
G-Retriever	1048 ✗	✗	50.0
LightRAG	1042 ✓	✗	41.7
LEGO	1042 ✓	✓	80.0

Baseline failure mode. Four of five baselines retrieve Civil Code Art. **1048** (prohibition of consanguineous marriage), which is *topically adjacent* (located in the same chapter as Art. 1042) but *doctrinally unrelated* to the case. This is a textbook near-neighbour retrieval miss: the embedding space confuses two provisions about “marriage prohibitions” of fundamentally different kinds. LightRAG retrieves the correct Art. 1042 but does not compose it with the invalidity-effects chain (Art. 157), and so its overall reasoning chain is incomplete.

LEGO’s contribution. LEGO is the only method that simultaneously retrieves the correct

Art. 1042 *and* the correct Art. 157 (effects of invalid juristic acts including fault-based loss allocation), and additionally cites Art. 964 (intermediary not entitled to fee for failed brokerage). The composition produces a three-step invalidity argument—*prohibition* → *invalidity* → *restitution with fault allocation*—that no baseline assembles. Reasoning component score is 0.75 versus ≤ 0.50 for all baselines; Law component is 0.75 versus 0.50 for baselines. The +21.7 Overall margin is the largest LEGO win in the PLawBench dataset and isolates two reinforcing mechanisms:

- **Near-neighbour avoidance.** ExpertGraph’s normative edges distinguish “prohibition by marriage-trade” from “prohibition by consanguinity,” allowing LEGO to bypass the lexical attraction to Art. 1048 that traps four baselines.
- **Multi-statute composition.** Syllogistic CoT explicitly chains Arts. 1042 → 157 → 964 along an invalidity-restitution path, rather than treating each article as an independent citation.

J.5 Cross-Case Synthesis for PLawBench-Civil

The four cases isolate four distinct baseline failure modes and the corresponding LEGO mechanisms:

Baseline failure mode	Example
Hedged conditional conclusion on bivalent question	civil-107
Doctrinally off-target anchor statute	civil-682
Missing controlling statute under noisy distractors	civil-56
Near-neighbour statute confusion (Art. 1048 vs. 1042)	civil-633

Across these cases, LEGO’s gains arise from (i) normative-edge retrieval that distinguishes lexically similar but doctrinally distinct provisions, (ii) element-by-element decomposition of the major-premise statute, and (iii) multi-statute composition along invalidity, restitution, and fault-allocation chains. 65

K LexRAG_Civil Case Studies

This appendix expands the LexRAG_Civil row of Table 2 into case-level analyses. Unlike PLawBench’s case-analysis format, LexRAG asks short consultation questions in the context of a multi-turn dialogue history, and judges responses on three dimensions (Overall, Factuality, UserSatisfaction; each $\in [1, 10]$).

ID	Category	LEGO	Best B.	Δ
972_turn2	Real-property Dispute	8	4 (LightRAG)	+4
173_turn4	Public Security	9	6 (HippoRAG 2)	+3
301_turn2	Inheritance	9	7 (HippoRAG 2)	+2
639_turn1	Land Dispute	9	6 (G-Retriever)	+3

Table 24: Four LexRAG_Civil case studies, drawn from the top of the LEGO-vs-best-baseline margin distribution.

On LexRAG_Civil the statistical picture differs sharply from PLawBench: LEGO’s gains over every baseline are *statistically significant* on paired-difference tests (all $|t| > 2$; $p < 0.05$ against HippoRAG 2 and $p < 0.001$ against the four remaining baselines). LEGO’s mean per-sample rank is 2.19 (vs. 3.27 for Naive RAG, the next best), and LEGO is uniquely best on 30/140 samples and tied for best on 31 additional samples.

Selection. We enumerate all 140 samples, rank by $\text{Overall}_{\text{LEGO}} - \max_b \text{Overall}_b$, and pick the four cases below to span four distinct consultation categories and four distinct baseline failure modes.

K.1 972_turn2 (Real Property): The Effect-of-Registration Doctrine

Question (turn 2 of 2). *Does failure to obtain a property certificate affect the transfer of real-property rights?*

Dialogue context. Turn 1 established that the user is asking about a so-called “triple-receipt house” (*sanlian-dan fang*)—a residential property held under informal documentation rather than a regular real-estate certificate.

Gold articles. Civil Code Arts. **209** (registration effects real-property right transfer) and **210** (registration procedures).

Reference answer. The establishment, modification, transfer, and extinction of real-property rights must be registered in accordance with law to take effect. If the property certificate has not been obtained, the actual transaction can form only a creditor’s right between the parties; no property-right transfer occurs.

Per-method scores and verdicts on the core legal question.

Method	O	F	S	Verdict
Naive RAG	3	2	3	“does not affect”—incorrect
RAPTOR	2	2	2	“does not affect”—incorrect
G-Retriever	3	2	2	internally contradictory
HippoRAG 2	2	2	2	“does not affect”—incorrect
LightRAG	4	3	3	“does not affect”—incorrect
LEGO	8	8	8	“generally does affect”—correct

Baseline failure mode. All five baselines commit the same doctrinal error: they conflate *registration as a condition of validity* for real-property right transfer (Art. 209, applicable here) with *registration as a condition for assertability against third parties* (Art. 225, applicable to specific movables such as vehicles and vessels). By writing “the absence of registration does not affect the validity of the property-right transfer, but merely deprives it of third-party assertability,” the baselines apply the wrong doctrine to real property—a classical textbook error reflecting embedding-space proximity between two structurally similar but legally distinct rules.

LEGO’s contribution. LEGO’s response correctly invokes Art. 209 and *separates the contract from the property right*:

If a housing sale-purchase contract has been signed but the transfer-of-title registration has not yet been completed, then although the contract itself is valid (under Art. 215), the property right (i.e., ownership of the house) has not actually been transferred to the buyer...

This is exactly the reference’s distinction between the creditor’s right arising from the contract and the property-right transfer requiring registration. LEGO is the only method that anchors the answer in the *principle of separation* (between the obligatory act and the dispositive act), which is the controlling doctrine here.

K.2 173_turn4 (Public Security): Joint Liability among Multiple Tortfeasors

Question (turn 4 of 4). *I wasn’t the only one in the fight—do I have to bear all the compensation alone?*

Dialogue context. Turns 1–3 established: a fight occurred, the user was hit first, the user sustained injury, and the assistant already explained Art. 234 (criminal assault), Art. 20 (self-defence), Art. 1173 (comparative fault), and Art. 1165 (general tort liability). The user is now narrowing the question to *multi-actor liability apportionment*.

Gold articles. Civil Code Arts. **1168** (joint tort yields joint and several liability) and **1170** (multi-person endangerment with possibly identifiable specific tortfeasor).

Statute retrieval and Overall.

Method	Cited	Overall
Naive RAG	1174 / 1175 (wrong: third-party / victim fault)	1
RAPTOR	973 (partnership debt!)	2
G-Retriever	178 / 518 (general joint liability)	3
HippoRAG 2	1170 (one of two gold)	6
LightRAG	1168 (the other gold)	3
LEGO	1170 + 178 (recourse)	9

Baseline failure mode. Three of five baselines retrieve *wrong* articles: Naive RAG goes to victim-fault rules, RAPTOR jumps to partnership liability (Art. 973 governs commercial partnerships and is irrelevant), G-Retriever picks a general joint-liability article without multi-tortfeasor specificity. Two methods (HippoRAG 2, LightRAG) each retrieve one of the two gold articles but cannot compose a complete answer.

LEGO’s contribution. LEGO’s response (i) directly answers the user’s framing (“you do not necessarily have to bear it all alone”), (ii) decomposes Art. 1170 into its two sub-cases—*identifiable specific tortfeasor* vs. *unidentifiable*—and (iii) closes with the *recourse right* under Art. 178 (a joint debtor who pays over their share may recover from co-debtors). LEGO also leverages dialogue history: turns 1–3 already cited Art. 1173 (comparative fault), so LEGO does not redundantly re-cite it and instead introduces the new doctrine (Art. 1170) the user actually needs. On multi-turn consultations, LEGO’s advantage is partly retrieval (avoiding wrong-statute traps) and partly conversational pragmatics (answering the newly-narrowed question rather than re-deriving the full multi-turn framework).

K.3 301_turn2 (Inheritance): Separating the Inheritance Act from Property Transfer

Question (turn 2 of 2). *Does subrogated inheritance require transfer-of-title procedures?*

Dialogue context. Turn 1 confirmed that the user is entitled to subrogated inheritance under Art. 1128 because her mother predeceased her grandmother.

Gold articles. Civil Code Arts. **1124** (acceptance of inheritance) and **208** (real-property registration).

Reference answer. Two layers must be separated: (i) inheritance acceptance is governed by Art. 1124—silence equals acceptance, no procedural act required; (ii) the subsequent real-property registration is governed by Art. 208 and *is* required for the heir to better exercise the right.

Method outputs and Overall.

Method	Answer to the core question	O
Naive RAG	flat “no”	3
RAPTOR	flat “no”	2
G-Retriever	hedged “not necessarily”	3
HippoRAG 2	distinguishes via Arts. 1128 + 230	7
LightRAG	flat “no”	2
LEGO	separates inheritance act from property transfer	9

Baseline failure mode. Three of five baselines give a flat “no” answer—technically correct for the inheritance *acceptance* act but misleading because the user is asking in the context of inheriting a real estate. The user’s practical question is *do I need to do paperwork?* A flat-no answer leaves her unaware that property-register transfer is still required for her to exercise ownership.

LEGO’s contribution. LEGO is the only method besides HippoRAG 2 to separate the two legal events the question conflates:

Subrogated inheritance itself does not require transfer-of-title procedures... In practice, however, if the estate involves real property (such as housing), transfer-of-title procedures are required to complete the change of ownership. This procedure is needed for the property-right transfer, not as a legal effect of subrogated inheritance itself.

LEGO additionally distinguishes *when the property right vests* (Art. 230: at the moment inheritance opens) from *when registration is procedurally required* (for the heir to dispose of the property). This mirrors the reference’s two-layer structure, and explains the Overall margin of +2 over HippoRAG 2 and +6 over the three flat-no baselines.

K.4 639_turn1 (Land Dispute): Indefinite-Term Lease of Private Land

Question (turn 1 of N). *If I lease my private land to a tenant and we do not specify a fixed term, what governs the lease?*

Gold articles. Civil Code Arts. **704** (required clauses of a lease) and **715** (termination rights for indefinite-term leases).

Reference answer. Three layers: (i) Art. 704 requires parties to include a lease term; (ii) under

Art. 715, either party may terminate with reasonable notice if no term is specified; (iii) practical advice—sign a supplementary written agreement.

The same two mechanisms account for both the rubric-based gains on PLawBench-Civil and the dialogue-quality gains on LexRAG_Civil.

Statute retrieval and Overall.

Method	Cited	Overall
Naive RAG	705 (20-yr cap)	5
RAPTOR	721 (rent-payment timing) — off-topic	4
G-Retriever	707 (writing requirement)	6
HippoRAG 2	707 + 510 + 734 (renewal)	5
LightRAG	705 (20-yr cap)	5
LEGO	707 + 730 (priority renewal) + 705	9

Baseline failure mode. Each baseline retrieves a single topically relevant provision but no method assembles the complete chain that the reference requires: form requirement, indefinite-term consequence, termination right, and priority renewal for the existing tenant. RAPTOR’s choice of Art. 721 (rent-payment timing) is a particularly clear retrieval failure—the lease chapter was located, but the wrong article within it was selected.

LEGO’s contribution. LEGO integrates three provisions (Arts. 707, 730, 705) and frames the answer as structured advice with explicit action items: sign a written contract, fix a term of ≤ 20 years, prioritize the existing tenant on renewal. The LexRAG UserSatisfaction dimension explicitly rewards actionable practical guidance, which only LEGO provides. When multiple provisions interact—form rule, default rule, and right-of-first-refusal—LEGO’s multi-hop retrieval composes them into a coherent advisory response; baselines retrieve one provision and present a partial answer.

K.5 Cross-Case Synthesis for LexRAG_Civil

The four LexRAG_Civil cases isolate four LEGO mechanisms, three of which echo the PLawBench findings (Appendix J) and one of which is specific to multi-turn consultation:

Mechanism	LexRAG case	PLawBench analogue
Near-neighbour doctrinal disambiguation (validity-registration vs. opposability-registration)	972_turn2	civil-633 (Art. 1042 vs. 1048)
Avoidance of wrong-statute retrieval traps	173_turn4	civil-56 (Art. 1177 vs. 1165)
Multi-layer concept disambiguation	301_turn2	civil-107 (joint debt + exception clause)
Multi-statute composition + dialogue-history awareness	639_turn1, 173_turn4	(new on LexRAG)

The mechanisms instantiate the two design choices of LEGO: ExpertGraph’s normative edges disambiguate provisions that are lexically close but doctrinally distinct, and Syllogistic CoT composes multiple provisions into structured advisory chains rather than presenting them as parallel citations.

L Statistical Validation of the Main Results

This appendix reports the paired significance analysis for the main results in Table 1 and for the ablation study in Table 3. The analysis is post hoc: it re-uses the stored per-item predictions and changes no prediction, score, or reported accuracy.

Procedure. All comparisons are paired at the item level over the $n = 723$ LawExamQA_Civil items. For each pair of systems we treat the two binary per-item correctness vectors as matched observations and apply a two-sided *exact* McNemar test, which conditions on the discordant pairs and so does not rely on a large-sample approximation. Confidence intervals are unadjusted percentile intervals obtained by resampling items with replacement 10,000 times and recomputing the paired accuracy difference on each resample; the resampling unit is the item, so pairing is preserved within every resample. Intervals describe effect-size uncertainty, while family-wise decisions use Holm-adjusted p -values.

Comparison families. We treat the system-level comparisons and the component-level ablations as two separate families and apply the Holm correction within each. The two families answer different questions – whether LEGO improves on externally published systems, and whether each of its own components contributes – and pooling them would penalise both for the size of the other. Family 1 is the thirteen comparisons of Table 1; family 2 is the eleven non-full configurations of Table 3.

System-level results. LEGO’s advantage over all five same-backbone RAG systems survives correction, with adjusted p between 1.5×10^{-9} and 4.4×10^{-6} and intervals well clear of zero. The margins over the larger general-purpose models do not: the intervals for Qwen3-30B-A3B, GPT-5 and DeepSeek-V3 all span zero, and the DeepSeek-V3 margin corresponds to four items out of 723. We scope the claim accordingly. The comparison with larger models is not evidence that an 8B backbone is intrinsically stronger; it indicates that expert-structured retrieval and syllogistic reasoning supply domain signal that lets a small open model reach the accuracy band of models one to two orders of magnitude larger. The statistically supported claim is the fixed-backbone one: under a shared Qwen3-8B reader and a shared Civil Code corpus, LEGO improves on every RAG pipeline we evaluate.

LEGO vs.	ΔAcc (pp)	95% CI	p_{Holm}
<i>Same-backbone RAG systems</i>			
Naive RAG	11.62	[+8.30, +15.08]	1.5×10^{-9}
LightRAG	11.76	[+8.16, +15.35]	5.1×10^{-9}
RAPTOR	11.20	[+7.75, +14.66]	8.4×10^{-9}
HippoRAG 2	10.65	[+7.19, +14.11]	3.4×10^{-8}
G-Retriever	8.85	[+5.39, +12.31]	4.4×10^{-6}
<i>Larger general models</i>			
Qwen3-30B-A3B	3.60	[−0.41, +7.47]	0.27
GPT-5	3.04	[−1.11, +7.05]	0.33
DeepSeek-V3	0.55	[−3.46, +4.56]	0.84

Table 25: System-level comparison family: paired accuracy differences between LEGO and other systems in Table 1, with exact McNemar tests, percentile bootstrap intervals (10,000 item-level resamples), and Holm correction applied across all thirteen comparisons in the family. Positive Δ favours LEGO. The eight rows listed cover every same-backbone RAG system together with the three closest larger models; the five comparisons not shown have larger margins.

Configuration	ΔAcc (pp)	95% CI	p_{Holm}
Naive RAG + plain CoT	6.23	[+2.49, +9.82]	0.005
ExpertGraphRAG + plain CoT	4.98	[+1.38, +8.58]	0.019
Naive RAG + ExpertCoT	4.70	[+1.52, +7.88]	0.019
ExpertGraphRAG + IRAC-CoT	3.18	[−0.14, +6.50]	0.067

Table 26: Component-level comparison family: accuracy drop of each non-full configuration relative to the full LEGO system, with Holm correction applied across all eleven non-full configurations of Table 3. Full LEGO is significantly better than ten of the eleven; the single exception is the closest configuration, ExpertGraphRAG + IRAC-CoT. The four rows listed are the three non-full cells of the 2×2 crossed design analysed below, plus that closest configuration.

Component-level results. Every ablation that removes an expert component costs accuracy, and ten of the eleven differences clear the corrected threshold. The single exception is instructive rather than damaging: ExpertGraphRAG + IRAC-CoT, the strongest non-full configuration, is 3.18 pp below the full system with an interval that just includes zero ($p_{\text{Holm}} = 0.067$). Once expert-structured retrieval is in place, the further benefit of replacing a strong generic legal prompt with ExpertCoT is positive in point estimate but not separable from noise at this sample size.

Interaction between the two modules. Two questions have to be kept apart: whether both modules materially contribute, and whether their combination establishes statistical super-additivity. The crossed ablation supports the first; the second re-

mains inconclusive. On the complete 2×2 subset $\{\text{Naive RAG}, \text{ExpertGraphRAG}\} \times \{\text{plain CoT}, \text{ExpertCoT}\}$, the conditional gains reinforce one another: the ExpertGraphRAG gain rises from +1.24 pp under plain CoT to +4.70 pp under ExpertCoT, and the ExpertCoT gain rises from +1.52 pp under Naive RAG to +4.98 pp under ExpertGraphRAG. The corresponding interaction point estimate is +3.46 pp. It is estimated from the four paired outcome vectors over all 723 items, not from a small subset of them, but its jointly paired-bootstrap 95% interval includes zero ($[-0.97, +7.75]$; 10,000 resamples). The results therefore support architectural integration and complementary conditional gains, and we make no formal claim of statistically established super-additivity.

Neither module is dominant. The crossed ablation also does not support reading ExpertGraphRAG as a modest retrieval add-on to the prompt. With ExpertCoT held fixed, replacing ExpertGraphRAG with Naive RAG reduces accuracy from 40.53% to 35.82% ($\Delta = 4.70$ pp, 95% CI $[+1.52, +7.88]$, $p_{\text{Holm}} = 0.019$). The factorial average marginal effects are 3.25 pp for ExpertCoT and 2.97 pp for ExpertGraphRAG; their 0.28 pp difference – algebraically the paired contrast between the two off-diagonal conditions – is not distinguishable from zero (paired-bootstrap 95% CI $[-3.60, +4.15]$; exact McNemar $p = 0.944$). The evidence does not identify either module as dominant. Functionally, ExpertGraphRAG raises Recall@8 over the Civil Code corpus from 27.99% to 72.53%, while ExpertCoT structures the application of the retrieved provisions.

Discordant-pair counts. Because the exact McNemar test conditions on discordant pairs, we report them for the two closest same-backbone comparisons. Against G-Retriever, LEGO is correct where G-Retriever is wrong on 116 items and wrong where G-Retriever is correct on 52, a net gain of 64 items; against Naive RAG the split is 129 against 45, a net gain of 84. These are the quantities the tests above rest on. They should not be confused with the twenty cases discussed in Appendix H, which are a deliberately stringent fourteen-system intersection – LEGO correct and all thirteen alternatives wrong – used for qualitative mechanism analysis rather than as a count of pairwise wins.